BANK OF THAILAND

Monetary Policy Report

March 2014

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2014

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Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprise of the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim to provide monetary policy decisions that are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy with the objective to support sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

On December 4, 2013, the MPC and the Minister of Finance signed a Memorandum of Understanding to continue using the quarterly average core inflation with the range of 0.5-3.0 percent as monetary policy target for 2014. However, due to the dissolution of parliament on December 9, 2013, approval of the proposed target must come from the new Cabinet which is yet to be established. In the meantime, the MPC shall continue to use the old monetary policy target until formal approval can be sought.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the key policy rate to signal the monetary policy stance.

Evaluating economic conditions and forecasts

The Bank of Thailand considers information from all sources, the macroeconomic model, data in each economic sector, as well as direct opinions from large enterprises and small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and covers all aspects, both in the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of Monetary Policy communication to the public, the MPC arrange many forms of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 hrs. on the day of the meeting, (2) publishing Minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

March 2014

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy appeared poised to slowdown in 2014 due to weaker domestic demand during the first half of the year. This was due to the political situation in Thailand which dented consumer and investor confidence. Meanwhile, exports gradually recovered in line with improved trading partners' economies. Nevertheless, should the political situation subside by mid-2014, domestic demand was expected to pick up and would be the driver of economic growth together with exports. As a consequence, the Thai economy would resume growing close to its normal pace in 2015. Meanwhile, inflationary pressure edged up mainly from the pass-through of LPG cost to food prices, while demand pressure softened in line with economic conditions.

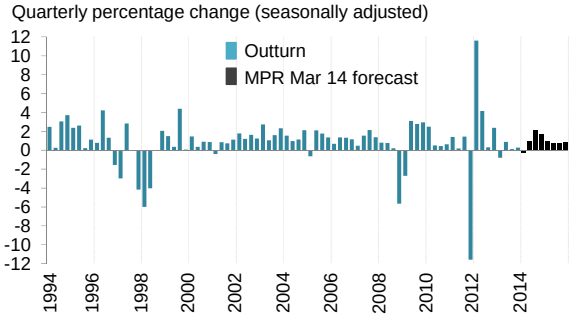
In the past three meetings, the MPC voted to reduce the policy rate by 0.25 percent in the first meeting, hold the policy rate at 2.25 percent per annum in the following meeting and then voted to reduce the policy rate by another 0.25 percent to stand at 2.00 percent per annum in the latest meeting. The MPC deemed that there was room for monetary policy to ease in order to lend more support to the economy during the recovery period, while inflationary pressure was not yet a concern.

1.1 Growth and inflation prospects

The Thai economy was likely to expand at a slow pace during the first half of 2014, while inflationary pressure edged up mainly from the pass-through of LPG costs to food prices.

In 2014, the Thai economy was expected to expand at a much slower pace than previously projected, especially in 2014 H1. This was due to lower-than-expected economic momentum in the latter half of 2013, from contractions in domestic demand and slower-than-expected export recovery. Moreover, uncertainty in

Chart 1.1 Thailand's Quarterly GDP growth



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted)

the Thai political situation which took place since end-2013 directly affected public spending and further weakened consumer, investor and tourist sentiment. As a result, domestic demand in the first half of the year was expected to grow at a slow pace. Nevertheless, gradual recovery in exports, owing to the pickup in trading partners' economies, would be the main engine of growth this year. The MPC assessed that **Thai economic growth should revert to its normal level in 2015 if the political situation subsided by midyear**. This would be supported by improvements in private spending and the tourism industry, the main engines of economic growth, together with continued recovery of the export sector, as well as public spending which would play a greater role in economic growth **(Chart 1.1 and Table 1.1)**.

Table 1.1 Forecast summary

Percent	2013*	2014	2015
GDP growth	2.9	2.7	4.8
Headline inflation	2.2	2.5	2.3
Core inflation	1.0	1.5	1.3

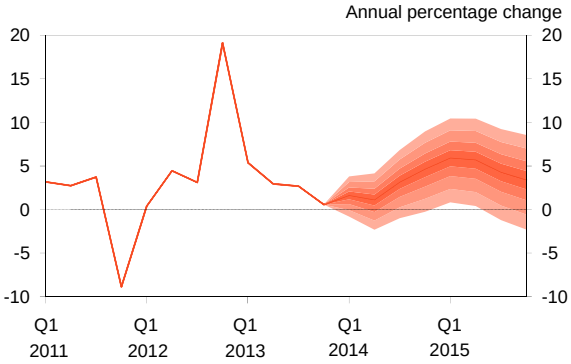
Note: * Outturn
Source: Office of the National Economic and Social Development Board and calculations by Bank of Thailand

Risks to the Thai economy stemmed from both domestic and external factors. Domestic risk factors tilted towards the downside if the political uncertainty prolong into the latter half of 2014. This could directly affect public spending by a delay of the 2015 budget process, while consumer, investor and foreign tourist confidence would also be further distressed. At the same time, external risk factors also tilted towards the downside, owing to consequences from the Fed's gradual tapering of

Quantitative Easing (QE) measures which could cause Emerging Market Economies (EMEs) to moderate by more than expected in the baseline assumption used by the MPC. This was in part a result of capital outflow and volatile exchange rates. In turn, the risk of economic slowdown in EMEs could delay recovery of Thai exports. On the whole, the MPC viewed that downside risks to economic growth remained high. The fan chart for growth was thus skewed down throughout the projection period **(Chart 1.2)**.

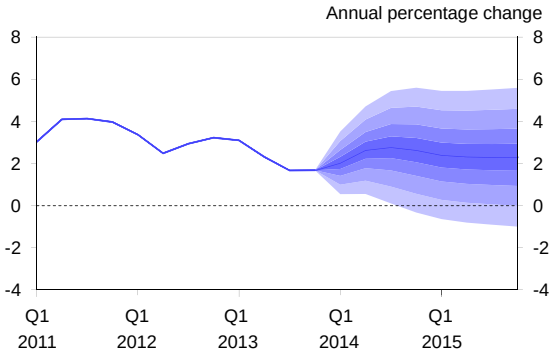
Inflationary pressure edged up from higher pass-through of household LPG cost to food prices than previously assessed, while demand pressure moderated in line with economic conditions. Steady hikes in household LPG price since September 2013 caused food prices to increase significantly, despite relatively stable prices of other goods following subdued demand pressure from the economic slowdown. Inflation projections for both headline and core inflation were thus adjusted up throughout the projection period, especially for 2014 **(Table 1.1)**. Furthermore, the MPC assessed that risks to inflation were balanced. Risk that inflation could be higher than projected stemmed from a higher-than-assessed pass-through of LPG cost to food prices. Meanwhile, risk that inflation could be lower than assessed could arise from lower-than-expected domestic demand should the economy grow slower than the baseline case **(Charts 1.3 and 1.4)**.

Chart 1.2 GDP growth forecast



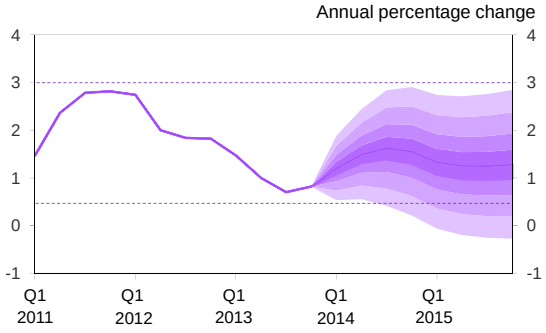
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.3 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.4 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

1.2 Economic outlook

Domestic demand was poised to grow at a slow pace in the first half of the year. Meanwhile, exports would play a greater role as driver of growth in line with the pickup in trading partners' economies. Moreover, if the political situation subsided within the middle of the year, the Thai economy should be able to resume normal growth rates in 2015.

The Thai economy was poised to moderate

In 2014, the Thai economy was expected to expand at a substantially slower pace than last anticipated, in part due to lower-than-previously assessed domestic demand, both private and public, in the latter half of 2013. Household consumption contracted in 2013 Q3, particularly in durable goods, following gradual waning of government stimulus measures and acceleration in household debt. Private investment also contracted as businesses postponed investments to assess the economic situation. Later on, political tension which started to build up since end-October 2013, put a further dent in private confidence. As a result, private spending recorded a larger contraction in the fourth quarter, which would have a continuous impact during the first half of 2014. Moreover, the political situation which prolonged and intensified periodically affected the tourism atmosphere, and was expected to reduce the number of foreign tourists in the first half of this year.

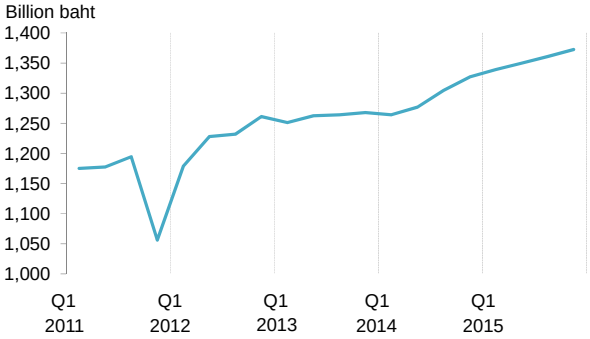
With regard to public spending, political uncertainty which began after the dissolution of parliament affected public spending, both budget and non-budget, particularly, non-budgetary

expenditures on water management along with infrastructure investment projects, which were previously expected to be disbursed this year.

Nevertheless, if the political situation was resolved by mid-2014, which was the baseline assumption used by the MPC, private demand would gradually recover to its normal trend in 2015. Household spending would improve in tandem with higher confidence in economic conditions and income. Meanwhile, businesses were likely to resume investments, especially those that were aimed to improve productivity and replace labor, which had been postponed from the previous 1-2 years. For the public sector, most of the 2014 budget was expected to be disbursed as usual, although some disbursements from the central budget and capital expenditure could be somewhat affected. Meanwhile, promulgation of the 2015 budget was expected to be delayed by one quarter. The tourism industry was poised to recover quickly from the political situation, similar to previous experience in which short recovery period was noted. For the export sector, although actual figures for 2013 reflected slower-than-expected export recovery, this year, the sector should benefit more from stronger economic recovery of trading partners and would then take on a greater role in supporting economic growth, particularly in the latter half of 2014.

Weaker-than-expected economic momentum towards the end of 2013 **(Chart 1.5 and Table 1.2)**, coupled with the expected slowdown in domestic demand in the first half of 2014 led to **significant downward adjustments in economic growth projections for 2014 (Table 1.1). For 2015, the MPC assessed that Thai economic growth was likely to resume its normal pace**,

Chart 1.5 Level of GDP



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted by the Bank of Thailand)

thanks to stronger domestic demand and exports, as well as public spending which would resume normal appropriations.

Table 1.2 Forecasts for GDP and components

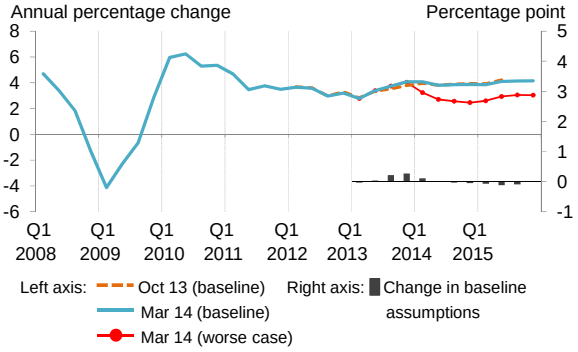
Percent	2014	2015
GDP growth	2.7	4.8
Domestic demand	0.6	5.3
Private consumption	0.3	4.6
Private investment	-0.5	10.2
Government consumption	2.9	2.4
Public investment	1.7	2.7
Exports of goods and services	4.5	7.7
Imports of goods and services	1.1	9.1

Note: At 1988 prices

The global economy recovered more concretely

The global economy picked up, led by major industrialized economies, in particular the U.S. and the euro area which broke out of recession. Meanwhile, the Chinese economy was poised to moderate from measures implemented to rebalance China's economic structure.

Chart 1.6 Growth assumptions for Thailand's trading partners



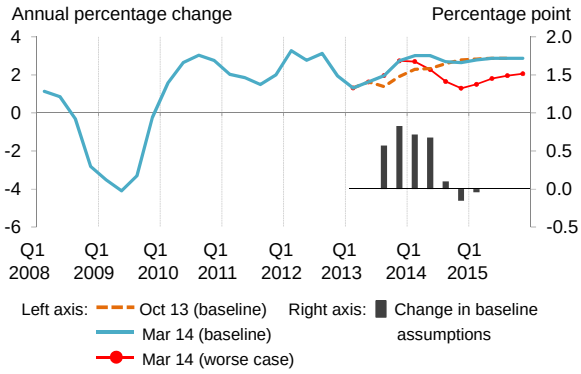
Note: Weighted by each country's share in Thailand's total exports

Thailand's trading partners' economies improved steadily. Momentum in 2014 remained the same as previously evaluated (Chart 1.6). The U.S. economy was expected to grow at a faster pace than previously projected, thanks to continuous improvements in domestic demand and the labor market. Euro area economies started to exhibit positive growth and were expected to recover slowly. Overall, Asian economies were expected to grow steadily in line with previous projections. Asian economies (excluding China)

grew slightly higher than expected, in particular countries that benefited from exports which improved in line with the global economy. Meanwhile, the Chinese economy slowed down somewhat due to implementation of measures to reduce overheating in public and private investment.

The U.S. economy was expected to pick up (Chart 1.7). Although financial conditions tightened somewhat from the rise in long-term interest rates after the Fed started to unwind its third round of quantitative easing (QE3) measures, economic recovery should not be stalled, thanks to steady improvements in U.S. economic fundamentals. The U.S. economy was expected to expand by 2.8 and 2.9 percent in 2014 and 2015, respectively. This was supported by stronger momentum in private consumption following continuous pickup in the labor market and businesses' stronger financial position. In addition, the fiscal drag[1/] and fiscal risks from the debt ceiling problem[2/] abated. Nevertheless, risks to U.S. economic recovery

Chart 1.7 Growth assumptions for the U.S.



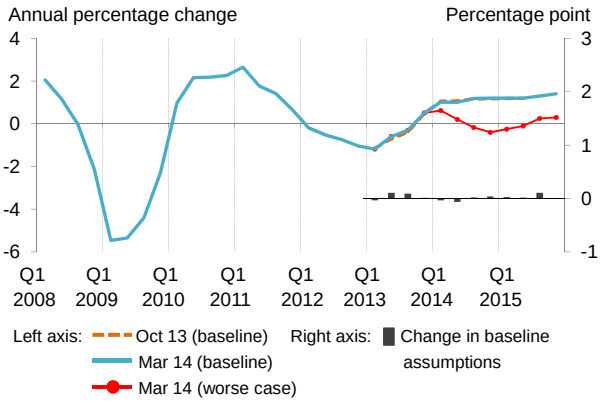
[1/] On December 18, 2013 the US Congress passed the 2014 and 2015 budget and eliminated some of the spending cuts required by the sequester in the 2014 and 2015 budget, which increased the central government budget by 45 and 18 billion US dollars, respectively. The spending cuts that were eliminated in 2014 and 2015 would be compensated by (1) an increase in revenue collection from tourism user fees and an increase in contribution into the pension fund, and (2) additional spending cuts on Medicare, all totaling 85 billion US dollars which would be spread out into the next 10 years.

[2/] On February 12, 2014 the US Congress passed a bill to suspend the federal debt ceiling until March 15, 2015. Within that period, the government would be able to borrow money, as appropriate, through the issuance of bonds. Then on March 15, 2015, when the suspension ends, the debt ceiling would be set equal to the actual debt level on that day.

remained, stemming from resolution of the sovereign debt problem which would take time and could weaken investor sentiment periodically.

Continuous improvements in economic fundamentals led the Fed to gradually unwind QE3 from a monthly purchase of 85 billion U.S. dollars to 75 billion U.S. dollars in January 2014 and reducing it further to 65 billion U.S. dollars in February 2014. The Fed also signaled that it would hold the fed funds rate at 0-0.25 percent per annum if the unemployment rate remained above 6.5 percent and inflation was below 2.5 percent.

Chart 1.8 Growth assumptions for the euro area



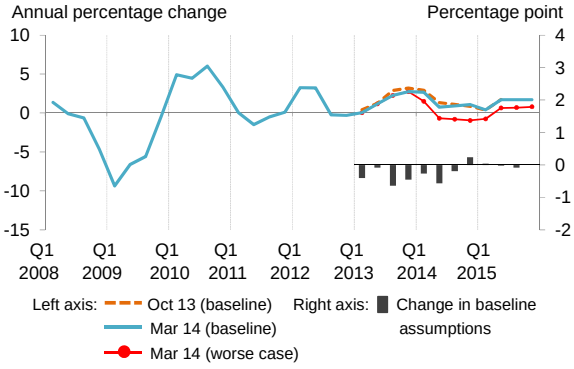
In 2014, euro area economies would resume positive growth, albeit at a tepid pace (Chart 1.8). The euro area exhibited a positive growth rate in the fourth quarter of last year, with recovery expecting to be at a slow pace. Such outlook was supported by lower fiscal drag and economic reforms, both fiscal and monetary, which started to show positive results, as well as more concrete plans regarding the establishment of the Banking Union which was set up to address problems in the banking sector.

Nevertheless, euro area economies remained fragile on the back of weak economic fundamentals, including record high unemployment rates in many countries and imbalances among countries in the group due to differences in the level of competitiveness. The main drag on euro area economies stemmed from 1) the deleveraging process of both private and public sectors; 2) structural problems in many countries which would take time to resolve, such as rigid labor markets and low competitiveness; and 3) slowdown in credit extension, especially to business sectors that had been contracting continuously. These factors would impede recovery of the private

sector. In addition, risk of deflation was expected to remain low, as inflation expectations indicated that inflation would adjust towards the central bank's target in the medium-term.

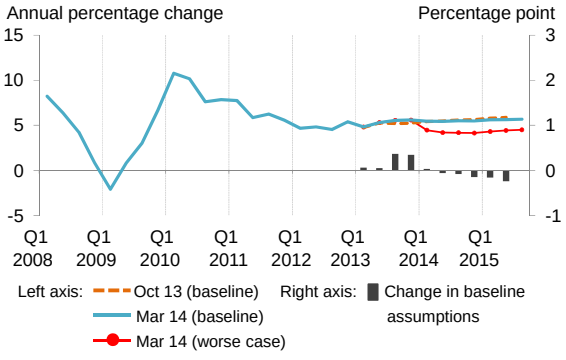
The Japanese economy was likely to lose momentum after the hike in consumption tax (Chart 1.9). In 2014, the Japanese economy was poised to moderate following the hike in consumption tax from 5 to 8 percent in April 2014. Momentum was likely to accelerate in 2014 Q1 before the tax hike, but this was expected to lose some thrust after the tax measure came into effect. Government stimulus measures could help alleviate impacts from the tax hike, but only to a certain degree. Meanwhile, the export sector was expected to pick up in line with trading partners' economic recovery and weaker yen. Nonetheless, uncertainties regarding impacts from the tax hike and supporting measures, as well as clarity in the economic structural reform plan could pose risks to economic recovery in Japan, going forward.

Chart 1.9 Growth assumptions for Japan



Asian economies (excluding Japan) were likely to expand continuously, from 5.5 percent in 2014 to 5.7 percent in 2015, driven mainly by the export sector (Chart 1.10). In particular, Asian countries (excluding China) would benefit from more broad-based exports which recovered in tandem with the global economy. Meanwhile, domestic demand in ASEAN economies was expected to moderate somewhat owing to private investment as financial institutions became more stringent in lending, especially to the real estate sector, and from a decline in the fiscal deficit.

Chart 1.10 Growth assumptions for Asian economies (excluding Japan)



Note: Weighted by each country's share in Thailand's total exports

The Chinese economy was expected to expand by 7.6 percent in 2014 and 2015. Although exports were likely to improve and would be the main factor supporting economic growth going forward, domestic demand, in particular investment which has a 48 percent share in GDP, was poised to weaken. The slowdown was in part a result of the Chinese authorities' intention to rebalance China's economic structure in order to achieve sustainable growth by reducing reliance on investment and increasing the role of consumption. In this regard, the Chinese government implemented measures to reduce the role of public investment, especially those of local governments and state enterprises, as well as reduce reliance on the shadow banking industry which had caused the financial sector to become more fragile in past periods. Measures to slow down loan growth to the real estate sector were also issued.

Exports would be the main engine of growth

Exports of goods were likely to recover in tandem with trading partners' economy. However, exports of services were affected by the political situation, but should recover quickly.

Merchandise exports were expected to edge up from last year (Table 1.3). Exports would increase from the previous year and play a greater role in supporting the economy. The recovery in exports was in line with expansions in trading partners' economies, especially G3 countries. These countries showed clear signs of recovery along with Asian economies which exhibited continued growth. In particular, Thai export goods that still bore a competitive edge were likely to expand well, such as automobile and parts, electrical appliances, computer and parts, processed agricultural products and fishery

products. The steady pickup in exports would help cushion the slowdown in the Thai economy in the first half of 2014. Additionally, clearer signs of export recovery in the latter half of 2014 would help boost recovery in private investment and speed up production to accommodate higher external demand. However, risks to export recovery remained from the shrimp supply shortage problem following the disease outbreak and deteriorating price competitiveness in rice exports. Another risk to export recovery was limitation of the export production structure such that Thai exporters were unable to produce high-tech products to keep pace with changing global demand, for example, the fast changing consumers' preference from personal computers which use hard disk drives to tablets and smart phones which use solid-state drives.

Exports of services in terms of tourist revenue were expected to grow at a slower pace than previously assessed. The outlook for the tourism industry in 2014 was affected by the ongoing political situation, in particular during the 60-day imposition of the Emergency Decree on January 22, 2014. This led many foreign governments to lift the level of travel warning for Thailand. However, tourist confidence should recover quickly after the Emergency Decree was lifted, while confidence in safety should increase once the political situation subsides. This was because Thailand's tourism industry still possessed distinctive strengths and potential. As a result, exports of services were likely to accelerate sharply together with exports of goods during the latter half of this year, and should continue along this expansionary trend into 2015.

The current account balance[3/] was likely to be close to balance in 2014, but would record a slight deficit in 2015 (Table 1.3). The current account balance was poised to improve from the 2013 deficit, trending close to balance in 2014. This was due to the gradual increase in exports of goods, while imports contracted in tandem with sluggish domestic demand. Nevertheless, the current account balance was expected to record a slight deficit in 2015 in response to steady increases in import demand for consumer goods, raw materials and intermediate goods, as well as capital goods and machinery, consistent with the recovery in domestic demand.

Table 1.3 Forecasts for the external sector

	2014	2015
Growth in value of exports* (F.O.B., percent)	4.5	6.5
Growth in value of imports* (F.O.B., percent)	3.0	7.5
Trade balance (billion U.S. dollars)*	9.8	8.2
Current account balance (billion U.S. dollars)*	0.7	-0.5

Note: *Data revision according to definitions in IMF's Balance of Payments Manual, 6th edition (BPM6), and Ministry of Commerce's revised database

Private demand should soften in the first half of the year

Private demand was likely to weaken more than expected due to a slowdown in household consumption and business investment, but was expected to pick up in the latter half of the year given that the political situation improves.

Private consumption in 2014 was likely to expand at a considerably lower rate than previously projected, in part due to lower-than-expected momentum of private spending in

3/ Including reinvested earnings

2013 H2. Accelerated private spending in previous periods, thanks to government stimulus measures along with reconstructions and repairs of flood-related damages, led to a longer-than-expected payback period of durable goods consumption. Moreover, households became more cautious in non-durable goods spending on the back of higher debt burden. In addition, the political situation further dented consumers' confidence in the economy and future income during the end of last year. The MPC assessed that if the political situation subsided by mid-year, private consumption would gradually pick up and gain momentum during the end of the year after effects from the payback was over and the spending environment improved. This would be bolstered by improvements in income and employment conditions following recovery in exports, which in turn would improve households' financial position and strengthen consumer confidence.

Private investment in 2014 was also expected to expand at a lower rate than projected in the previous *Report*, particularly in the first half of the year. This was partly due to investments in 2013 H2 which was much lower than previously assessed, coupled with weaker private confidence owing to the prolonged political uncertainty since the end of last year. As a result, investments in machinery and equipment contracted, while manufacturers delayed new investments to assess the economic and political situation. In addition, the construction companies deferred investments following a cool down in the real estate market.

The MPC assessed that private investment was likely to recover slowly in the first half of the year due to the political situation which extended into 2014 H1 and sluggish recovery in the export sector. Private investment would mainly be for necessary replacement and maintenance of machinery and equipment. Meanwhile, investment in construction could also continue to moderate amidst uncertainty in consumers' purchasing decisions. Nonetheless, given that the political situation subsides, starting from the latter half of this year new investments in machinery and equipment to enhance productivity, as well as openings of new construction projects could pick up speed in accordance with original investment plans that had previously been postponed. This pickup would be supported by improved confidence in domestic consumers' purchasing power and the export sector which recovered in line with the global economy, as well as more concrete economic management by the government.

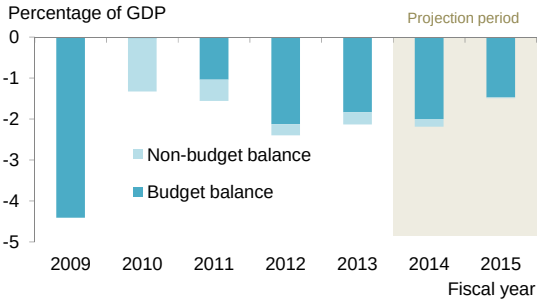
Nevertheless, with regard to private investment in the medium- to long-term, the MPC viewed that the business sector should give greater importance in improvement of production potential and enhancement of production technology to keep pace with the changing global demand. Moreover, businesses should reduce reliance on labor in order to accommodate labor shortage and low labor productivity problems, as well as reduce transportation and logistics costs which would require support from the public sector in terms of infrastructure investment.

Fiscal stimulus declined from the previous assessment

Fiscal stimulus declined, in particular investment expenditures on large projects. Overall, this year's fiscal stimulus would be similar in size to last year's.

Fiscal policy in 2015 would continue to provide economic stimulus, but at a lesser extent than evaluated in the previous *Monetary Policy Report* (Chart 1.11). In fiscal year 2014, the fiscal deficit was expected to stand at 2.2 percent of GDP, similar to last year's level.

Chart 1.11 Assumptions on public sector expenditure



Note: Non-budgetary spending includes (1) the Mega Investment Project (Strong Thailand), (2) the water management project, and (3) the Development Policy Loan (DPL)
Source: Public Debt Management Office and Bureau of the Budget and calculation by Bank of Thailand

The political situation which stretched into 2014 after the dissolution of parliament on December 9, 2013, directly affected both budgetary and non-budgetary spending. The disbursement rate for the 2014 budget was expected to stand at 90.5 percent, lower than the previous assessment, on the back of lower disbursements from both current and capital expenditures. Furthermore, political factors were likely to delay the 2015 budget process by one quarter. However, the government should be able to expedite budget disbursements for the remainder of the fiscal year, given the setup of the Budget Disbursement Committee which oversees the expedition of budget disbursements as was practised in the past **(more details in BOX: Dissolution of parliament: the impact on budget disbursement)**.

Assumptions on public spending were adjusted downwards throughout the projection period (Table 1.4), in particular the central budget for emergency and essential projects. With regard to non-budgetary expenditures, especially for the water management project, disbursements would be possible only for projects that required

immediate attention and could be performed by authorities, for example, canal dredging and road repairs. Meanwhile, only some projects under the infrastructure investment plan were expected to be implemented, in particular investment projects that were ready and continued from the government's original investment projects. Furthermore, the political situation also delayed disbursements of transfers and allocation of funds to local governments, as well as economic spending plans. Postponement of some investments by state enterprises would also take place.

Table 1.4 Assumptions on public sector expenditure

Unit: Billion baht	Fiscal year	
	2014	2015
General government consumption	1,668.3	1,741.5
Public investment	686.6	722.2
Total	2,354.9	2,463.7

Note: Includes expenditure assumptions on the water management project and the infrastructure investment project
Source: Bureau of the Budget and calculation by Bank of Thailand

Inflationary pressure edged up from the previous assessment

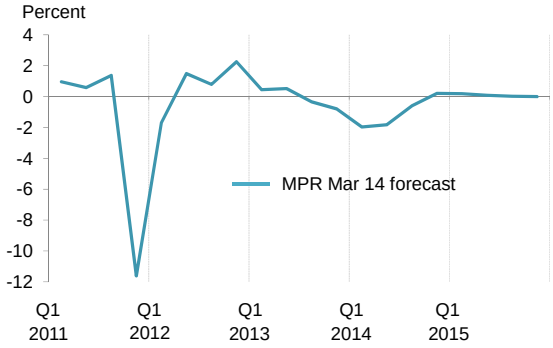
Inflation was likely to increase, owing largely to the pass-through of household LPG cost to prepared food prices. Meanwhile, other commodity prices were expected to remain stable given that demand pressure and crude oil prices were poised to decline.

Food inflation was higher than previously projected. The gradual increase in the household LPG price since September 2013 of 50 satang per kilogram per month led prepared food prices to rise steadily since end-2013 to 2014 Q1, and was higher than previously assessed. As a result, core inflation and headline inflation in the first quarter

was substantially higher than projected. Looking ahead, prepared food prices would continue to increase in tandem with hikes in the LPG price that would continue until September 2014. Nevertheless, the extent of pass-through was expected to decline from previous periods as the economic slowdown could make the pass-through of costs to prices more difficult.

Inflationary pressure from the demand side was likely to ease in line with the weakening economy. Softer domestic demand, especially during the first half of the year following high uncertainty in the economic and political situation, coupled with high debt burden in some households, led the household sector to spend more cautiously. The weakening economy caused the negative output gap to increase in the first half of the year **(Chart 1.12)**. However, once the economy started to pick up in the latter half of the year, the output gap would begin to narrow and move closer to zero, which could prompt demand pressure to increase somewhat in 2015.

Chart 1.12 Output Gap



Cost pressure from world crude oil and non-fuel commodity prices were poised to decline in 2014. The MPC assessed that, going forward, **Dubai crude oil prices** would be lower than previously appraised, remaining at 105.0 U.S. dollars per barrel throughout the projection period **(Chart 1.13)**. The downward adjustment was supported by easing concerns over short-term tightening of oil supply which stemmed from unrest in oil producer countries in the Middle East and Africa, such as Syria and Libya, which had subsided. Moreover, long-term oil supply was expected to increase following higher crude oil production in non-OPEC countries, particularly in North America which should be sufficient to accommodate recovery in global demand. The MPC thus deemed that supply and demand of

Chart 1.13 Assumptions on Dubai oil price

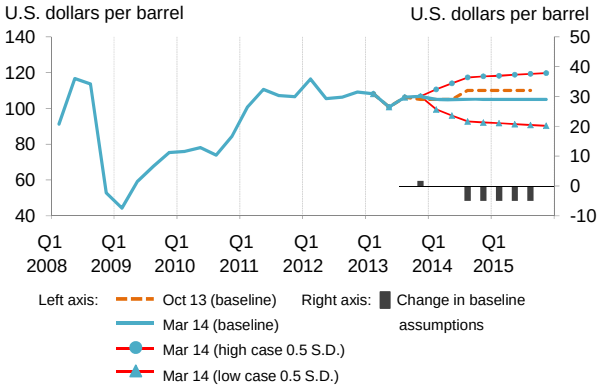


Chart 1.14 Assumptions on non-fuel commodity prices

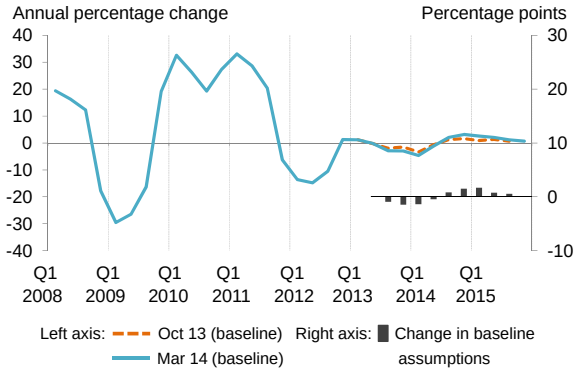
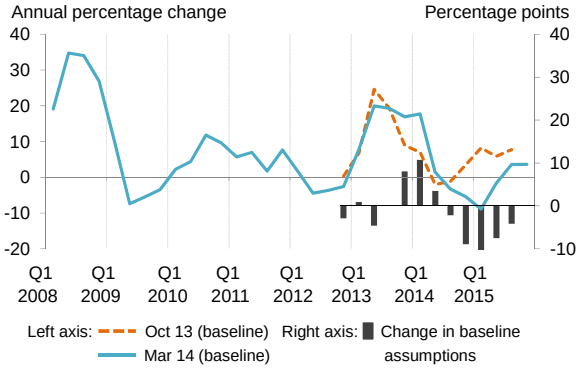


Chart 1.15 Assumptions on fresh food prices



world crude oil would likely balance out throughout the projection period. As a result, the high case and low case assumptions on the Dubai oil price were balanced at half a standard deviation above and below the baseline path. Meanwhile, **world nonfuel commodity prices** declined **(Chart 1.14)** owing mainly to prices of metal and agricultural raw materials, such as rubber and cotton, which fell in line with slower-than-expected recovery in global demand, especially from China, in the first half of the year. At the same time, stocks remained high. Meanwhile, prices of food and beverages remained elevated due to drought conditions in Brazil.

For 2014, **domestic fresh food prices** adjusted up from the previous assumption following price increases in meat and agricultural products (excluding rice), especially shrimp price which rose higher than expected owing to low supply from the disease outbreak, while external demand increased in tandem with global economic recovery. At the same time, the decline in rice price this year was consistent with the increase in supply due to expedited rice sales from the government's rice pledging scheme. In addition, global rice price was likely to decline given high world production and stock of rice, such as in Indonesia. For 2015, the MPC assessed that domestic fresh food prices should trend downwards following the expected increase in shrimp production after the disease outbreak situation improved **(Chart 1.15)**.

Furthermore, given the government's policy to control domestic energy prices, cost pressure from the transport sector was expected to remain low, declining from the previous assessment. The MPC maintained the assumption of a diesel excise tax exemption throughout the projection period. The MPC also

assumed that the original plan to increase LPG price for the transport sector, beginning in March 2014, would be postponed. This was to adhere to the decision of the Committee on Energy Policy Administration (CEPA) that future policies would be under the discretion of the new government.

Looking ahead, the MPC assessed that risks to inflation would be relatively balanced, with upside risk stemming from higher pass-through of household LPG cost to food prices than assessed in the baseline case as the gradual hike in LPG prices of 50 satang per kilogram per month would continue until September 2014 along with the observation that underestimation of pass-through of LPG cost to food prices had been common in previous periods. On the other hand, the downside risk to inflation would stem from lower-than-expected demand pressure should the economy slow down by more than projected in the baseline case.

Alternative scenarios

The MPC appraised that there were various scenarios where Thai economic growth could deviate from the baseline case. The MPC thus took these scenarios into consideration to reflect key risk factors on the Thai economy and inflationary pressure, going forward.

Scenario 1: Uncertainty in the domestic political situation extended into the latter half of 2015. This would affect private and tourist confidence, which in turn would cause private consumption and investment, as well as the tourism industry to recover slower than assessed in the baseline case. Moreover, financial positions of low income households would be more fragile, in particular those with high debt burden. The 2015 budget process could also be delayed by

more than one quarter and disbursements of the government's infrastructure investment projects could be limited further. Under this scenario, Thailand's economic growth would be lower than assessed in the baseline case throughout the projection period.

Scenario 2: Emerging market economies (EMEs) slowed down more than expected, due to impacts from the Fed's QE tapering. Risks from QE tapering throughout the year could cause capital outflow, as well as weaker and more volatile exchange rates. Consequently, EMEs could slow down by more than expected in the baseline case. Furthermore, QE tapering could also tighten liquidity in financial systems and reduce asset prices in EMEs. Nonetheless, sound external stability in Thailand, including high international reserves, should help the economy weather direct effects from capital flow volatility **(more details in BOX: Central banks' policy response to volatility in global financial markets and implications for Thailand)**. However, Thailand could be faced with indirect impacts from an economic slowdown in ASEAN countries and China, the country's major trading partner which accounted for 38 percent of export market in 2013. As a result, export recovery could be delayed and Thai economic growth could be lower than assessed in the baseline case.

Scenario 3: The Thai political situation subsided earlier than assessed in the baseline case. Under this scenario, private demand should start to pick up in the second quarter of this year, with respect to improvements in private sector confidence and the tourism industry which returned to normal. Consequently, the Thai economy would expand slightly higher than the baseline case.

1.3 Monetary policy decision

Monetary policy eased further

In the past three meetings, the MPC voted to reduce the policy rate by 0.25 percent in the first meeting, hold the policy rate at 2.25 percent per annum in the next meeting, and reduce the policy rate by another 0.25 percent to stand at 2.00 percent per annum in the latest meeting. The MPC deemed that there was room for monetary policy to ease further in order to lend more support to the economy during the recovery period which would take time.

In its meeting on November 27, 2013, the MPC assessed that the global economy would continue to recover; led by expansions in G3 countries, especially the U.S. Nevertheless, high uncertainty in the outlook for monetary and fiscal policies in the U.S. would continue to weigh on global financial market stability. The Chinese economy expanded well across all sectors, while regional economies continued to grow gradually. Meanwhile, the Thai economy expanded less than expected in 2013 Q3 from both private and public spending, while recovery in exports of goods had not gained traction. Furthermore, the Thai economy was projected to grow at a slower pace than previously assessed on the back of greater downside risks which stemmed from further delays in public investment, weakening private confidence from the political situation, and exports that had not yet recovered fully. Given benign inflation outlook and moderating household credit growth, there was room for monetary policy to mitigate downside risks to the economy and support economic recovery. The MPC thus voted 6 to 1 to reduce the policy rate by 0.25 percent, from 2.50 to 2.25 percent per

annum, with one member voted to hold the policy rate, deeming current monetary policy stance to be sufficiently accommodative.

In its subsequent meeting on January 22, 2014, the MPC assessed that G3 economies showed signs of gradual improvement, led by the U.S. economy which expanded on the back of stronger domestic demand. The Chinese economy expanded steadily, while exports of Asian economies, particularly in North Asia, began to recover. The Thai economy in 2013 Q4 was expected to grow less than previously assessed as a result of the ongoing political situation which put a dent in private confidence, weighing on overall outlook for growth. Meanwhile, exports continued to expand at a subdued pace. The MPC deemed that the current stance of monetary policy was accommodative and supportive of economic recovery. The ongoing political situation posed risks to growth, but sound economic fundamentals should help the economy weather these short-term risks. In addition, safeguarding financial stability remained a cornerstone for economic recovery in the periods ahead. The MPC thus voted 4 to 3 to maintain the policy rate at 2.25 percent per annum. Three members voted to reduce the policy rate by 0.25 percent as they viewed that monetary policy still had room to ease in order to guard against higher downside risks to growth.

With regards to the meeting on March 12, 2014, the MPC assessed that the global economy continued to recover, led by major economies. The Chinese economy expanded steadily, despite some signs of moderation in investment. Asian economies, overall, benefited from exports. Thai economic growth slowed in 2013 Q4 and January 2014 from contractions in domestic demand, with

private spending taking a hit by weaker private confidence. Meanwhile, tourism had begun to experience greater impact from the political situation. Going forward, political uncertainties would continue to impede recoveries of private consumption and investment. Nonetheless, exports of goods should gradually improve on the back of a recovery in major economies, which should provide impetus to growth this year. The MPC judged that downside risks to growth had risen in the wake of the ongoing political situation. Although core inflation edged up, it remained subdued. The MPC thus voted 4 to 3 to reduce the policy rate by 0.25 percent, from 2.25 to 2.00 percent per annum. Three members voted to hold the policy rate, viewing that the monetary policy stance was already accommodative while the main headwinds to growth were not financial in nature. Hence monetary policy should be used when it would be effective in supporting economic recovery.

1.4 Appendix:

Table 1.5 Forecast assumptions			
	2013	2014	2015
Dubai oil price (U.S. dollars per barrel)	105.4	105.0	105.0
Non-fuel commodity prices (%YoY)	-1.1	-0.3	1.8
Fresh food prices (%YoY)	16.0	2.2	-1.0
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (%YoY)[1/]	6.4	3.5	2.9
Public investment (%YoY)[1/]	0.8	6.6	6.0
Fed Funds rate (% at year-end)	0.13	0.13	0.75
Trading partners' economic growth (%YoY)[2/]	3.5	3.9	4.1
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	129.9	133.8	127.3

Note: [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each country's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by a decrease

Table 1.6 GDP growth forecasts by research houses		
	2014	2015
FPO[1/]	3.5-4.5	n.a.
NESDB[2/]	3.0-4.0	n.a.
DBS Bank	3.5	n.a.
HSBC	3.5	4.5
Standard Chartered	3.5	6.0
JP Morgan	3.0	4.2
Barclays	3.0	4.0
Kasikorn Bank	3.0	4.0
Kiatnakin Bank	2.8	5.0
Phatra Securities	2.8	5.0
BOT	2.7	4.8
Capital Economics	2.5	4.0
TISCO Securities	2.5	4.8
Siam Commercial Bank	2.4	4.5

Note: Compiled and published by Reuters on March 11, 2014, except:
[1/] Published on December 26, 2013
[2/] Published on February 17, 2014, with the release of GDP data for 2013 Q4
Presented in descending order of 2014's forecast

Table 1.7 Headline inflation forecasts by research houses		
	2014	2015
DBS Bank	2.9	n.a.
JP Morgan	2.6	n.a.
TISCO Securities	2.6	2.7
BOT	2.5	2.3
Capital Economics	2.5	3.0
Barclays	2.4	2.5
FPO[1/]	1.9-2.9	n.a.
NESDB[2/]	1.9-2.9	n.a.
Kasikorn Bank	2.4	2.8
Kiatnakin Bank	2.4	2.6
Phatra Securities	2.4	2.6
Standard Chartered	2.4	3.0
Siam Commercial Bank	2.2	2.4
HSBC	2.1	3.1

Note: Compiled and published by Reuters on March 11, 2014, except:
[1/] Published on December 26, 2013
[2/] Published on February 17, 2014, with the release of GDP data for 2013 Q4
Presented in descending order of 2014's forecast

Table 1.8 Probability distribution of GDP growth forecast								
Percent	2014				2015			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 12	0	0	0	0	1	1	0	0
10-12	0	0	0	2	5	5	2	1
8-10	0	0	1	8	15	14	8	5
6-8	0	0	8	19	25	23	17	13
4-6	3	5	24	27	26	25	24	21
2-4	33	24	32	24	17	18	23	24
0-2	49	39	23	14	8	9	15	19
(-2)-0	14	24	9	5	2	3	7	11
< (-2)	1	7	2	1	1	1	3	6

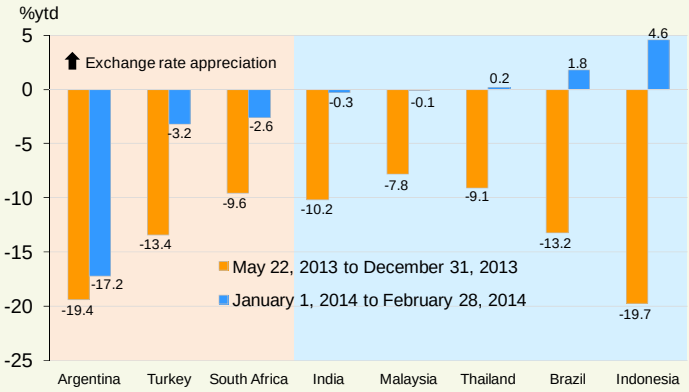
Table 1.9 Probability distribution of headline inflation forecast								
Percent	2014				2015			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 7	0	0	0	1	1	1	1	1
6-7	0	0	2	2	2	2	2	2
5-6	0	3	6	6	5	5	5	6
4-5	1	11	14	13	11	11	11	11
3-4	13	25	22	20	18	17	17	17
2-3	37	31	24	22	21	21	20	20
1-2	36	21	18	18	19	19	19	18
0-1	11	8	9	11	13	13	13	13
(-1)-0	1	2	3	5	6	7	7	8
< (-1)	0	0	1	2	3	4	4	5

Table 1.10 Probability distribution of core inflation forecast								
Percent	2014				2015			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
4.0-4.5	0	0	0	0	0	0	0	0
3.5-4.0	0	0	1	1	0	0	1	1
3.0-3.5	0	0	3	3	2	2	2	3
2.5-3.0	0	4	9	9	6	6	6	7
2.0-2.5	3	15	19	17	13	12	12	13
1.5-2.0	21	31	26	23	21	19	19	18
1.0-1.5	46	31	23	22	23	22	21	21
0.5-1.0	26	15	13	15	18	19	19	18
0.0-0.5	4	4	5	7	10	12	12	12
< 0.0	0	0	1	3	6	8	9	9

Central banks' policy response to volatility in global financial markets and implications for Thailand

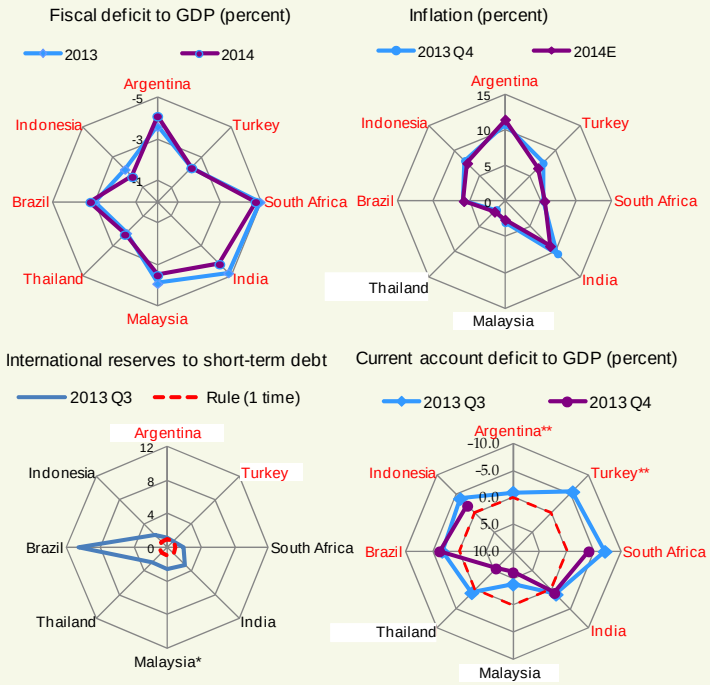
After the Fed signaled that it would start tapering QE in May 2013, emerging market economies (EMEs) experienced continuous outflows of capital. The capital flight was a result of asset sell-offs and the reallocation of funds into safe haven assets of major industrialized countries, causing EMEs' currencies to depreciate **(Chart 1)**.

Chart 1 Exchange rates of emerging market economies



Source: Bloomberg

Chart 2 Economic fundamentals of emerging market economies



Note: E = Estimates by the IMF
* Data for 2013 Q2
** Latest data until 2013 Q3

Source: CEIC, IMF, World Bank, Ministry of Finance Malaysia, Central banks and Bank of Thailand

Nevertheless, starting in early 2014, the currencies of EMEs started to move in diverse directions as investors began to give greater importance to economic fundamentals of each country. The extent of economic fundamental fragility, both domestic and external, helped differentiate risks of investing between various countries' assets. Thus, countries with higher fragility would ultimately experience higher risk of capital flight. After studying the latest figures of EMEs' fragile five[1/] and some ASEAN countries' economic fundamentals **(Chart 2)**, the followings were found. (1) there were high risks to domestic stability in some countries, in particular fiscal stability which weakened on the

[1/] The fragile five, which consists of Brazil, India, Indonesia, South Africa and Turkey, are countries judged to possess economic stability problems and are at risk of capital flight.

back of persistent budget deficits that were employed to boost the economy after the 2008 crisis, especially in South Africa, India and Argentina. Moreover, some countries were affected by high inflation, particularly in Argentina and India. (2) Although external stability had improved overall, risks still remained. Most countries were able to maintain sufficient international reserves to cover short-term external debt. Nevertheless loss in competitiveness and accelerated imports in past periods led to high current account deficits in Turkey and South Africa.

The difference in economic fundamentals was partly a result of differences in each country's economic structure. For example, Malaysian and Indian exports which benefited from the global economic recovery were important factors influencing the increase in Malaysia's current account surplus and the decline in India's current account deficit. Nonetheless, economic stability measures that were gradually implemented by some countries to withstand impacts from the Fed's tapering of QE were also important factors which helped boost investors' confidence and partly helped improve economic stability, resulting in positive impacts on EMEs' exchange rates' directions. Important economic stability measures implemented since 2013 by EMEs can be summarized as follows **(Table 1)**.

Table 1 Economic stability measures implemented by various countries

Year	India	Malaysia	Brazil	Indonesia	Argentina	Turkey	South Africa
2013	• Increase interest rate by 50 bps • Set up an FX swap line for commercial banks • Raise taxes/limit gold imports • Expand BSA limit with other central banks • Limit investments abroad • Reduce fuel subsidy	• Ration investment projects with high imports • Increase competitiveness • Reduce fuel subsidy • Abolish sugar subsidy	• Increase interest rate by 275 bps • Terminate collection of capital gain tax on investments in debt and equity instruments	• Increase interest rate by 175 bps • Reduce maximum L/D ratio • Raise RRR on rupiah deposit • Reduce holding period for SBI • Relax rules on FX purchases by exporters • Expand BSA limit with other central banks • Reduce fuel subsidy			
2014	• Increase interest rate by 25 bps	• Reduce subsidy on electricity • Reduce expenditures of government agencies	• Increase interest rate by 75 bps		• Limit purchase of imported goods	• Increase interest rate by 550 bps • Request commercial banks to increase RRR in local currency	• Increase interest rate by 50 bps

Note: BSA stands for Bilateral Swap Agreement
SBI stands for Bank Indonesia Certificates

1. **Measures to resolve high inflation issues and prevent capital flight.** Many countries resolved these problems by increasing the policy interest rate. Brazil continued to raise its policy interest rate gradually since April 2013, while Indonesia and India started hiking up their policy interest rates in mid-2013. After the interest rate hikes, downward trends of inflation were observed in these countries. At the same time, their currencies also became more stable. Meanwhile, Turkey and South Africa began increasing their policy interest rates in early 2014, with the hikes being quite substantial in size. Nevertheless, impacts of the policy interest rate hikes on inflation and capital outflows were not yet clear.

2. **Measures to reduce persistently high fiscal deficits**. Malaysia was committed to reduce the size of its budget deficit every year, aiming to achieve a balanced budget by 2020. In addition, Malaysia also implemented measures to steadily reduce public spending, which included cutting fuel and electricity subsidies, terminating sugar subsidies and reducing expenditures of government agencies. Similarly, apart from aiming to lower the budget deficit in 2015 from that in 2014, authorities in India and Indonesia also implemented measures to reduce public spending, such as curbing subsidies on fuel prices.

3. **Measures to cut down the current account deficit** differed between countries, depending on the specific cause of the deficit. For instance, India implemented measures to limit imports of gold due to high gold imports in past periods. In Indonesia, fuel subsidy cuts were not only aimed at reducing the budget deficit, but also at cutting down domestic fuel usage and fuel imports. Meanwhile, Malaysia also considered postponing investment projects with high import share but low economic impact to curb capital imports.

From the above experiences, it was observed that economic stability, both domestic and external, were important indicators of risk build-up. Countries with stronger economic fundamentals or ability to adjust more quickly would be able to better withstand impacts from external factors. Therefore, EMEs, including Thailand, must maintain strong economic fundamentals and take serious steps to prepare themselves for adverse impacts in order to shore up investors' confidence and stand ready to cope with volatility in the global financial market which could lead to sudden reversals of capital flows.

Recent Economic Developments

2. Recent Economic Developments

The global economy continued to recover in line with development in major industrialized economies, especially the U.S. where consistent improvement in fundamentals was observed. Growth in the euro area finally returned to positive territory, mainly owing to recovery in major economies. However, the Japanese economy moderated during 2013 H2 despite accelerated spending towards the end of the year in anticipation of an increase in consumption tax in 2014 Q2. Meanwhile, the outlook for Asia and China continued to be favorable thanks chiefly to improvement in exports.

The Thai economy in 2013 H2 slowed down from the same period last year owing to domestic demand contraction which stemmed from more cautious spending and investment behavior of the private sector. Domestic demand was also affected by the politcal situation coupled with delays in some parts of public investment after the dissolution of parliament. Such development in domestic demand coupled with the slow recovery in merchandise exports thus led to contraction in manufacturing production. In spite of this, tourism remained an important driver of growth, leading to strong expansion in tourism-related activities in the trade and service sector.

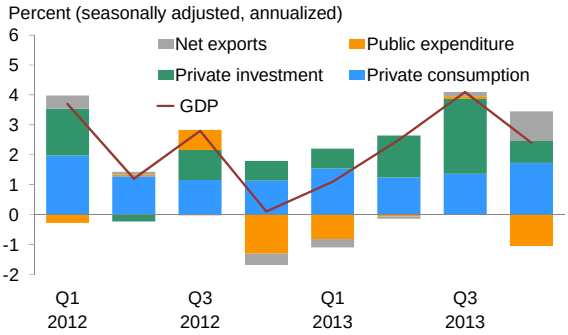
2.1 The global economy

The global economy continued to recover in line with development in major industrialized economies while exports from Asia and China benefited from improvement in these large economies more visibly.

During the second half of 2013, the U.S. economy recovered strongly thanks to continuous improvement in fundamentals, especially the labor and housing markets, despite uncertainties over its monetary and fiscal policies.

In the second half of 2013, the U.S. economy expanded on average by 3.3 percent each quarter (qoq, saar), accelerating from the

Chart 2.1 Contribution to the U.S.' GDP growth
(change from the previous quarter)



Source: Bureau of Economic Analysis

first half of the year in which growth averaged at 1.8 percent each quarter (qoq, saar), chiefly owing to continuous improvements in the labor and housing markets. **(Chart 2.1)** Such development led to stronger financial conditions of households and helped support consumption despite rising uncertainties from 1) tightened monetary conditions resulting from investors' expectation of QE tapering in May prior to the Fed's announcement of a reduction in asset purchase from 85 billion U.S. dollars per month to 75 billion U.S. dollars per month at the FOMC meeting in December 2013[1/]; 2) fiscal problem concerning the budget which resulted in a government shutdown on October 2013 for a total of 16 days; and 3) default risk of the U.S. government that could stem from restrictions on public debt ceiling.[2/]

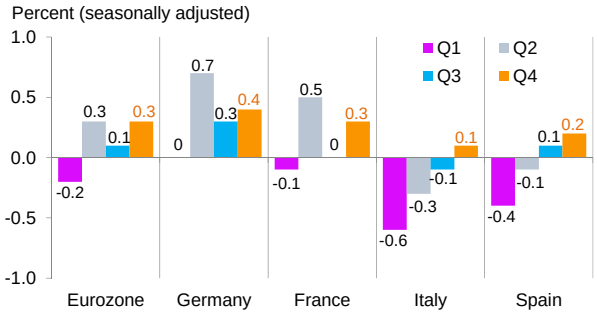
[1/] At the meeting on December 17–18, 2013, the FOMC decided to (1) reduce asset purchase by 10 billion U.S. dollars per month, resulting in a reduction in purchases of agency mortgage-backed securities from 40 billion U.S. dollars per month to 35 billion U.S. dollars per month, and will add to its holding of longer-term Treasury securities at a pace of $40 billion per month rather than $45 billion per month; and (2) maintain the fed funds rate at 0–0.25 percent per annum where this low interest rate environment would be maintained should the unemployment rate continue to be higher than 6.5 percent and inflation does not exceed 2.5 percent. The FOMC also signaled that it would maintain the fed funds rate at this level even if the unemployment rate moved below 6.5 percent, especially if inflation continued to be below the 2 percent target.

[2/] On October 17, 2013, the U.S. government agreed to extend the time period to pass the temporary budget and suspended enforcement of the requirements pertaining to the debt ceiling until January 17 and February 7, 2014, respectively.

The euro area[3/] recovered steadily during the second half of 2013 and while some progresses on fiscal tightening and establishment of a Banking Union[4/] were made, more time would be needed for a fully functioning Banking Union to be set up.

During the second half of 2013, the euro area continued to improve after pulling out of recession in the second quarter **(Chart 2.2)**. In 2013 Q3, economic growth was mainly driven by domestic demand while exports moderated owing to slowdowns in trading partners' economies, particularly Asian economies. However, exports recovered and became supportive of growth again in 2013 Q4, thanks to rising foreign demand. In addition, there were some progresses on economic reform and fiscal deficit reduction, particularly in Spain where efforts were spent to increase flexibility of the labor market and reduce spending associated with the pension system. In this connection, the euro-zone economy contracted by 0.4 percent for the whole year, improving from last year where a contraction of 0.7 percent was observed.

Chart 2.2 GDP Growth in the euro area in 2013
(change from the previous quarter)

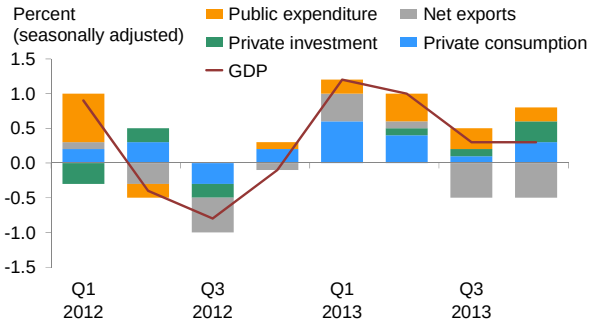


Source: Eurostat

[3/] The euro area consists of 17 European Union countries which have adopted the euro as their currency. In 2013, Germany, France, Italy and Spain each has a share in the overall Eurozone economy of 28, 21, 17 and 11 percent, respectively. Meanwhile Greece, Ireland and Portugal together hold 6 percent share in the overall Eurozone economy.

[4/] The Banking Union consists of three main mechanisms, namely 1) Single Supervisory Mechanism (SSM); 2) Single Resolution Mechanism (SRM); and 3) Deposit Guarantee Mechanism (DGM).

Chart 2.3 Contribution to Japan's GDP growth
(change from the previous quarter)



Source: Cabinet Office of Japan

In the second half of 2013, the Japanese economy moderated from the previous half despite accelerated household spending prior to the planned increase in consumption tax in April 2014.

In 2013 Q3, the Japanese economy continued to improve **(Chart 2.3)** albeit at a slower pace compared to the first half of the year. Such moderation was due to weakening private consumption following accelerated spending in prior periods owing to the government's stimulus measures, uncertainties surrounding the increase in consumption tax along with lack of clarity on the stimulus package to compensate the tax increase, causing households to postpone spending to future periods. Furthermore, exports also contracted following slowdowns in Asian trading partners' economies.

Nonetheless, after the government announced an increase in consumption tax from 5 percent to 8 percent at the beginning of 2013 Q4, to be implemented in April 2014, consumption accelerated again as consumers took advantage of the pre-consumption tax period. Moreover, manufacturing production and private investment also expanded in tandem with rising domestic and foreign demand.

The Chinese economy was poised to continue expanding mainly from improvement in exports. Asian economies excluding China also continued to grow in line also with exports and strong domestic demand, particularly North Asia.

In the second half of the year, the Chinese economy expanded on average by 7.8 percent, accelerating from 7.6 percent in the first half, mainly owing to continuous improvement in exports in line with recovery in the global

economy. Meanwhile, domestic demand, particularly investment, signaled some moderation **(Table 2.1)**, partly due to the authority's more cautious stance concerning risks in the financial sector and local governments. In this connection, the authority stepped up regulation in various economic sectors, for example, (1) measures to safeguard the real estate sector; (2) supervision of lending by the shadow banking sector; and (3) credit limits for local governments' asset management companies.

Table 2.1 Indicators of China's economic conditions (2013-2014)
(change from the same period the previous year)

	2013			2014	
	Q3	Q4	Dec	Jan-Feb	
				Actual	Forecast**
Investment in permanent assets*	20.3	19.7	19.7	17.9	19.4
Retail sales	13.4	13.5	13.6	11.8	13.5
Manufacturing Production Index	10.1	10.0	9.7	8.6	9.5
Exports	3.9	7.4	4.3	-1.7	3.8
Inflation	2.8	2.9	2.5	2.3	2.3

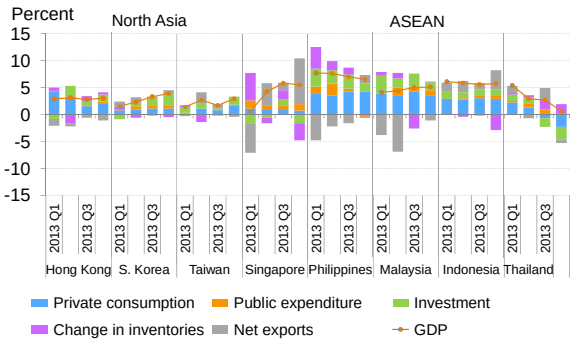
Notes: * From the beginning of the year until present
** Average forecast of exports and inflation during Jan-Feb
Sources: CEIC and Bloomberg

On March 5, 2014, the Chinese authority announced that economic growth target for 2014 would be maintained at 7.5 percent as was the case in 2013. **(Table 2.2)** Consumption growth target was also maintained at 14.5 percent while investment growth target was decreased from 18.0 percent to 17.5 percent, reflecting the authority's commitment towards rebalancing the economy by reducing dependency on investment and increasing the role of consumption on growth. In this regard, the government continued to pay attention to both economic reform and growth, resorting to both proactive fiscal policies along with prudent monetary policies.

Table 2.2 China's economic targets in 2013-2014 (change from the same period the previous year)		
	2013	2014
Economic growth	7.5	7.5
Investment in permanent assets*	18.0	17.5
Retail sales	14.5	14.5
Exports	8.0	7.5

Source: 2014 Annual meeting of the National People's Congress

Chart 2.4 Contribution to Asia's GDP growth
(change from the previous quarter)

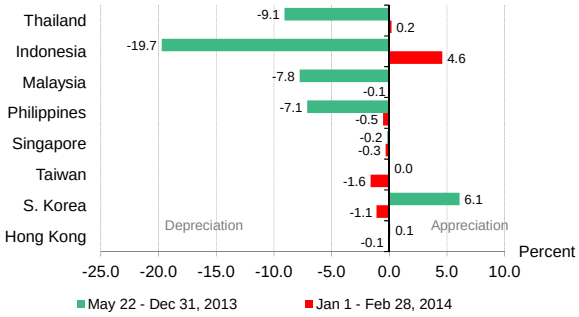


Source: CEIC

The economies of Asia excluding China continued to improve in the second half of the year **(Chart 2.4)** with exports in North Asia recovering quicker than those in ASEAN. This observation could be attributed to the fact that North Asia's exports consisted of medium- and high-tech goods as well as products that were crucial inputs in the global supply chains which greatly benefited from recovery in demand from advanced economies as well as China. Domestic demand that had been the main driver of growth in North Asian economies also continued to grow strongly while that in ASEAN began to show signs of weakness, owing to moderation in fiscal spending in efforts to reduce budget deficits, for example, the abolishment of oil subsidies in Malaysia and Indonesia. In addition, private investment also slowed down in part because financial institutions became more stringent in lending to some economic sectors, especially the real estate sector.

In the second half of 2013, Asia excluding China also faced with larger outflows of capital as a result of the Fed's announcement that it might begin to reduce asset purchase known as QE tapering, along with signs of moderation in regional economies. In this connection, foreign investors started to sell off assets in attempt to rebalance the portfolios towards major industrialized economies, which subsequently led to depreciation of Asian currencies. However, Asian currencies regained stability in early 2014, in part owing to improvement in economic fundamentals as well as continuous implementation of macroprudential measures to deal with the effect of QE tapering by some countries. **(Chart 2.5)**

Chart 2.5 Movements in Asian currencies



Source: Bloomberg

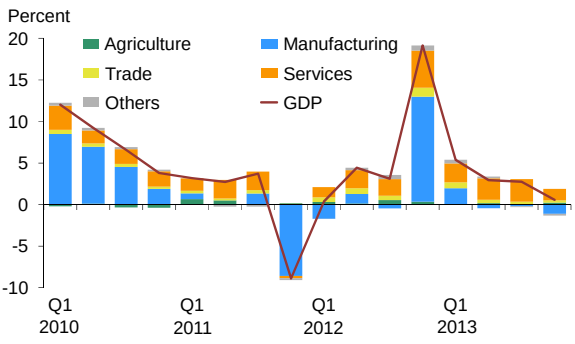
2.2 The domestic economy

In the second half of 2013, the Thai economy moderated in line with contraction in domestic demand and slow recovery in foreign demand. Subsequently, manufacturing production contracted. Nonetheless, the service sector continued to grow strongly, providing an important thrust for the economy.

Economic conditions in the second half of 2013

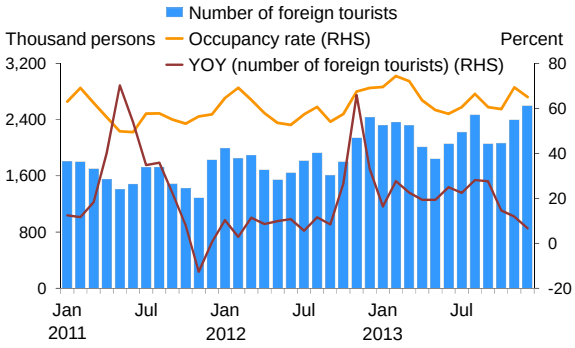
In the second half of 2013, the Thai economy expanded by 1.6 percent from the same period last year **(Chart 2.6)** thanks to tourism-related activities in the trade the service sectors along with increases in agricultural outputs. On the supply side, manufacturing production contracted in tandem with domestic demand slowdown following accelerated spending in prior periods.

Chart 2.6 Contribution to GDP growth
(change from the same period the previous year)



Source: Office of the National Economic and Social Development Board

Chart 2.7 Number of foreign tourists and hotel occupancy rate



Source: Department of Tourism and Bank of Thailand

Meanwhile, export-oriented manufacturing production was unable to provide compensational effect given the sluggish recovery in foreign demand coupled with supply constraints in some industries.

The **service sector** grew by 5.8 percent in the second half of 2013 and acted as the main driver of economic growth, particularly tourism-related industries such as transportation and hotel and restaurant industries which expanded highly in 2013 Q3 following an increase in the number of tourists from all regions including Europe, the U.S., and East Asia. **(Chart 2.7)** Nevertheless, the services sector began to moderate in 2013 Q4 in line with the tourism sector, partly because of a decrease in the number of Chinese tours after the Chinese authority introduced a law on quality control of tour packages in October. Moreover, travelers' concerns regarding the political demonstrations towards the end of the year also heightened, thereby causing the number of tourists from almost all regions to decline. However, the aforementioned impact was not as severe as during the political unrest in 2010 since some tourists were able to re-direct travel itineraries to other regions of the country more easily thanks to increases in air flights.

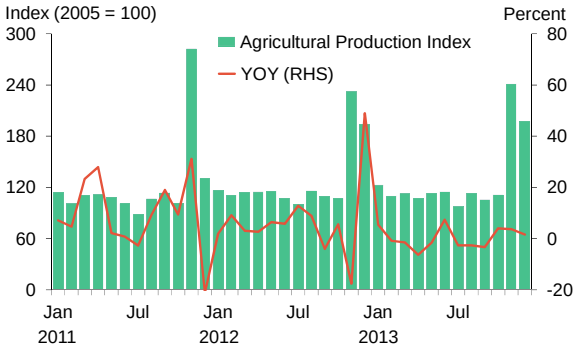
The **trade sector** moderated during the second half of the year owing to contraction in manufacturing production in some sectors along with slowdown in imports. Moreover, the construction sector contracted by 4.6 percent from the same period last year in line with slowdown in real estate activities and contractions in public and private investments which owed to the economic slowdown and delays in the government's large infrastructure investment projects. However, some construction by state enterprises was proceeded with, for example, construction of the purple and

blue MRT lines and construction of electrical power plants.

Agricultural production in the second half of 2013 expanded by 1.3 percent from the same period last year **(Chart 2.8)** in line with production of rice that grew well during the end of the year from favorable weather conditions together with rubber and oil palm production which rose from increases in plantation areas in previous periods. At the same time, shrimp production improved slowly towards the end of the year as shrimp farmers in some areas were able to gradually adjust production process to alleviate the impact of the outbreak of Early Mortality Syndrome (EMS).

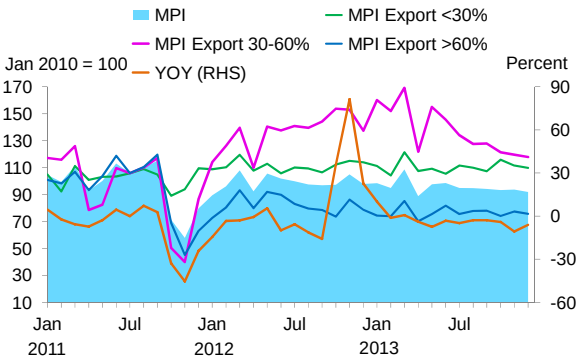
Manufacturing was the main sector which was affected by the slowdown in domestic demand. In 2013 H2, manufacturing production contracted by 1.7 percent from the same period last year **(Chart 2.9)**, where contractions were observed in both the third and fourth quarters. As such, capital utilization declined from 66.6 percent in 2013 H2 to 63.0 percent. **(Chart 2.10)** Domestic-oriented manufacturing production contracted, particularly that of automobiles and electrical appliances owing to significant reduction in demand for these goods, attributed in part by accelerated spending by households and businesses in prior periods together with the expiration of the first car tax rebate scheme and electrical appliance coupons. Moreover, increase in debt burden and deterioration in consumer and business sentiments owing to the political situation towards the end of 2013 Q4, led households to become even more cautious in their spending. Meanwhile, export-oriented manufacturing production continued to contract in both quarters, partly due to delays in economic recovery in major trading

Chart 2.8 Agricultural production



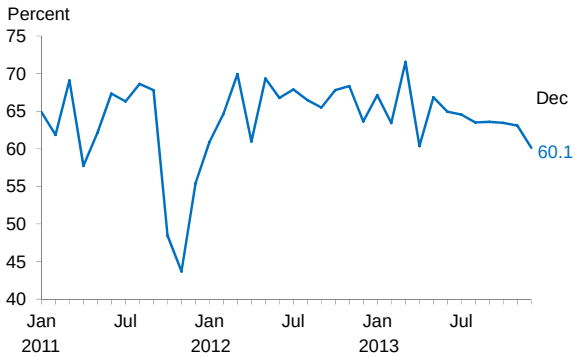
Sources: Office of Agricultural Economics and Bank of Thailand

Chart 2.9 Manufacturing Production Index



Source: Office of Industrial Economics, Ministry of Industry

Chart 2.10 Capital utilization rate



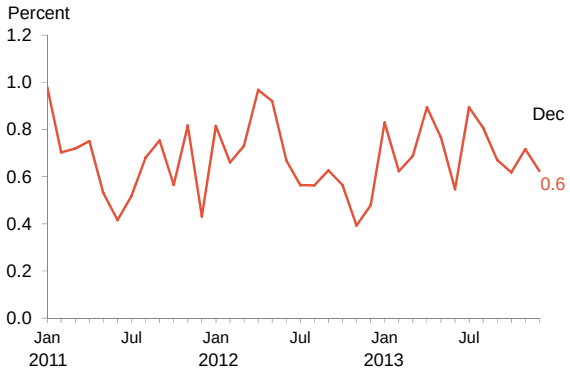
Source: Office of Industrial Economics, Ministry of Industry

partners, especially in 2013 Q3. Another factor was that there were supply constraints in some industries, which made businesses unable to fully benefit from strong improvement in the global prospect during 2013 Q4. In particular, the frozen shrimp industry was unable to meet the demand for exports in spite of gradual improvement in the EMS outbreak. Meanwhile, the frozen poultry industry could not operate at full capacity given that a major producer was unable to completely resume production after facing a liquidity problem. The hard disk drive (HDD) industry also faced a decrease in global demand as it could not adjust production to deal with changes in consumers' preference. Together, all these factors led to persistent contraction in Thai exports.

Nevertheless, manufacturing production in certain industries noted significant recovery in 2013 Q4 following improvement in the global outlook, particularly production of electronic tubes and parts. Some other industries also benefited from special factors which resulted in production increases towards the end of the year, including the petroleum industry in which demand edged up as a result of closures of oil refineries for maintenance in certain oil-importing countries along with increases in domestic oil stocks in accordance with the requirement of the Energy Policy and Planning Office. The HDD industry was another example where production increased thanks to requirements of HDDs as inputs for a new game console.

The economic slowdown during the second half of the year did not affect employment as reflected by the unemployment rate which continued to stabilize at 0.7 percent **(Chart 2.11)**. Overall, the corporate sector still had the need to employ as reflected by information from the BOT's

Chart 2.11 Unemployment rate



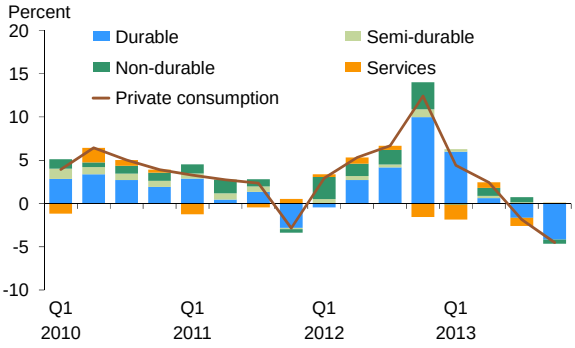
Source: National Statistics Office

business liaison program which found that businesses still faced with labor shortage and planned to hire more workers, going forward. Thus, Thailand's strong labor sector was expected to help absorb the impact of negative factors to private consumption in the periods ahead.

In the second half of 2013, domestic demand contracted from the same period last year due to slowdowns in private spending and public investment. Meanwhile, exports improved slowly in line with gradual recovery in the global prospect. As a result, domestic-oriented manufacturing production decreased from the first half of the year while export-oriented manufacturing production somewhat edged up slightly at the end of the year.

Private consumption contracted from the first half of the year **(Table 2.3)** in line with continuous decrease in demand for durable goods, especially automobiles, after some consumers had already accelerated this part of household spending in prior periods to take advantage of the first car tax rebate scheme. In this regard, the aforementioned accelerated spending, the increase in household debt along with financial institutions' more stringent lending to households, caused consumers to be more careful in their spending which ultimately cause consumption of non-durable goods to slowdown in 2013 Q3 and to finally contract in 2013 Q4 when consumer confidence began to deteriorate owing to the domestic political situation. **(Chart 2.12)**

Chart 2.12 Contribution to private consumption
(change from the same period the previous year)



Source: National Economic and Social Development Board

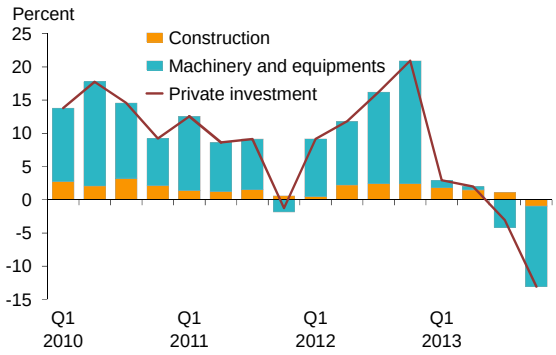
Table 2.3 GDP growth rate

Change from the same period in the previous year (percent)	2013				
	Year	Q1	Q2	Q3	Q4
GDP	**2.9**	**5.4**	**2.9**	**2.7**	**0.6**
Domestic demand	0.2	4.6	3.6	-1.4	-5.7
Private consumption	0.2	4.4	2.5	-1.2	-4.5
Private investment	-2.8	2.9	2.0	-3.1	-13.1
Government consumption	4.9	2.9	7.6	7.3	0.9
Pubic investment	1.3	18.8	15.4	-16.2	-4.7
Exports of goods and services	4.2	8.3	2.9	3.8	2.0
Goods	0.2	3.7	-1.5	-1.4	0.2
Services	19.7	25.7	22.4	25.2	8.0
Imports of goods and services	2.3	8.1	4.5	0.8	-3.5
Goods	1.8	9.5	4.4	-0.5	-5.3
Services	5.2	1.2	5.3	8.1	6.3

Note: 1/ Domestic demand excluding changes in stocks

Source: National Economic and Social Development Board and calculations by Bank of Thailand

Chart 2.13 Contribution to private investment
(change from the same period the previous year)



Source: National Economic and Social Development Board

Private investment also contracted from the first half of the year **(Chart 2.13)** as investment in machinery and equipment declined following accelerated spending in prior periods which owed to reconstructions and repairs after the flood. Moreover, contraction in domestic demand, sluggish recovery in exports, along with uncertainties from the domestic political situation caused some businesses to postpone investment to future periods in order to await greater economic and political clarity. At the same time, investment in construction began to exhibit some slowdown in 2013 Q3 after it had accelerated highly in the first half of the year. A contraction was later noted in 2013 Q4 owing to the economic slowdown which started to affect consumers' purchasing power, along with financial institutions' more stringent lending to real estate developers.

Exports improved slowly in the second half of the year thanks to more promising signs of recovery in 2013 Q4 after a contraction was witnessed in 2013 Q3 as exports of fisheries products were affected by a supply shock from outbreak of EMS in shrimp. Moreover, recovery in demand from major industrialized economies was also concentrated only in some products which were not Thailand's main export goods, for example, electronic goods such as tablets and smart phones, in which Thai manufacturers had yet to make technological adjustment to support their production and exports.

In 2013 Q4, prospect for exports began to brighten thanks to exports of agricultural products, electronics, electrical appliances, processed agricultural products, and petroleum products which started to benefit from recovery in major industrialized economies and China more visibly. It should be noted that exports of services remained Thailand's important source of growth given the continuous increase in the number of tourists. In this connection, annual growth rates in the number of tourists registered as high as 26.1 and 10.7 percent in 2013 Q3 and Q4, respectively, notwithstanding some slowdown in the last quarter of the year due chiefly to a decline in the number of Chinese tourists following the amendment in Chinese law to prohibit below-cost tour packages, which resulted in price increases. Additionally, the effect of Thailand's political situation on the tourism sector which became more evident towards the end of the year also contributed to the slowdown in the number of tourists. **(Chart 2.14)**

The contraction in domestic spending coupled with the slow recovery in exports subsequently caused imports to decline **(Chart 2.15)**, particularly raw materials and intermediate

Chart 2.14 Contribution to growth of exports
(change from the same period in the previous year)



Source: National Economic and Social Development Board

Chart 2.15 Contribution to growth of imports
(change from the same period in the previous year)

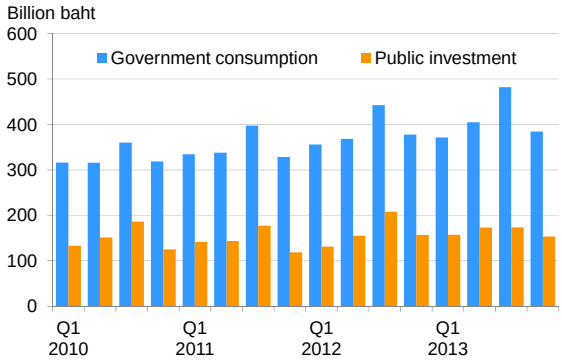


Source: National Economic and Social Development Board

goods (excluding crude oil) and machinery and equipment which were in part due to high base effects that resulted from accelerated imports for post-flood reconstructions and repairs in the previous year. Imports of non-durable goods also contracted along with those of durable goods in line with weakening private consumption.

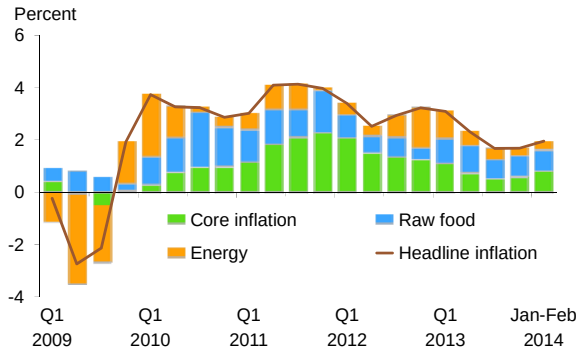
Public expenditure played a smaller role in contributing to growth compared to previous periods. In 2013 H2, government spending moderated from the same period last year **(Chart 2.16)** due to delays in disbursement of capital expenditure. Moreover, the impact of the political situation together with the dissolution of parliament at the end of the year also led to delays in disbursement of some parts of the central budget along with capital expenditure. Likewise, non-budgetary spending was mostly held back. However, disbursement of current expenditure, including salaries and purchases of goods and services could still be done as usual. Overall revenue collection declined from the same period last year, particularly excise tax on automobiles due to accelerated sales in the previous year owing to the government's first car tax rebate scheme, as well as import tax which decreased in line with import value.

Chart 2.16 Public expenditure



Source: National Economic and Social Development Board

Chart 2.17 Contribution to headline inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand

2.3 Costs and prices

Inflation

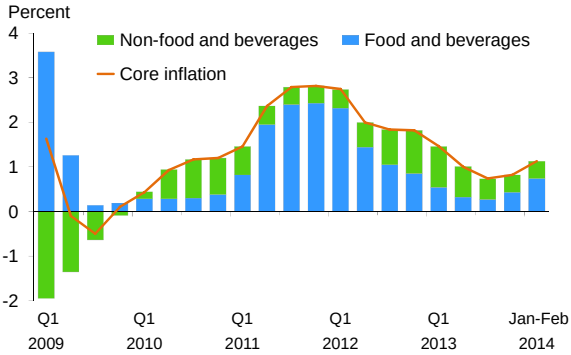
Headline inflation over the first two months of 2014 edged up from 2013 Q4 to register at 1.95 percent, owing to acceleration in core inflation and energy prices. **(Chart 2.17)**

Core inflation continued to increase from 0.82 percent in 2013 Q4 to 1.13 percent during the first two months of 2013. **(Chart 2.18)** In this regard, 90 percent of the increase in core inflation could be attributed by higher prices of prepared food, resulting from gradually rising pass-through from increases in costs of LPG and sauces and condiments since 2013 Q4.

Prices of energy accelerated from the previous quarter following an upward adjustment in the FT for the January – April 2014 round by 5 satang per unit, along with increases in domestic retail oil prices owing to higher cost of imports, resulting from the weakening baht against U.S. dollar.

Prices of raw food also accelerated from the previous quarter. Such development was explained by higher demand for meat consumption during the Chinese New Year season coupled with a meat supply shortage caused by severe drought.

Chart 2.18 Contribution to core inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce, and calculations by Bank of Thailand

Table 2.4 Quarterly inflation

Unit: Percent	2012	2013	2013				2014
			Q1	Q2	Q3	Q4	Jan-Feb
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index	3.02	2.18	3.09	2.32	1.67	1.68	1.95
• Core Consumer Price Index	2.09	1.00	1.47	1.00	0.74	0.82	1.13
• Raw food	4.40	5.54	6.08	6.60	4.08	5.04	5.17
• Energy	7.09	4.79	8.76	4.60	3.67	2.35	2.69
Percentage change from previous quarter (%Δqoq)							
- Headline Consumer Price Index	-	-	0.5	0.5	0.3	0.4	0.7
• Core Consumer Price Index	-	-	0.2	0.1	0.2	0.3	0.5
• Raw food	-	-	0.4	3.6	0.1	0.8	0.4
• Energy	-	-	2.1	-1.5	1.7	0.0	2.2
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index	-	-	0.4	0.03	0.16	0.34	0.22
• Core Consumer Price Index	-	-	0.2	-0.01	0.09	0.20	0.09
• Raw food	-	-	1.6	0.25	0.03	0.66	0.75
• Energy	-	-	-1.1	-0.05	1.22	0.50	-0.98

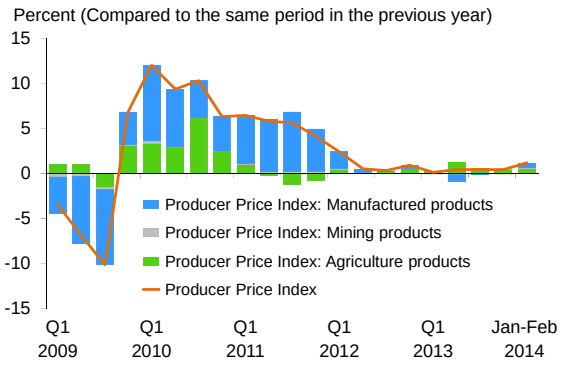
Source: Trade and Economic Index Bureau, Ministry of Commerce, and seasonal adjusted quarter-on-quarter percentage change calculations by Bank of Thailand

In this connection, headline inflation during the first two months of the year exceeded expectation owing to greater-than-expected input costs that is passed through to prepared food price. Energy prices were also underestimated due to higher-than-expected retail oil prices mainly attributed to the weakening baht against U.S. dollar. **(Table 2.4)**

Production cost conditions

During the first two months of 2014, the Producer Price Index expanded by 1.2 percent, edging up slightly from 0.4 percent in 2013 Q4 **(Chart 2.19)** in line with rising prices of petroleum products which owed to higher cost of imports that resulted from the weakening baht. Nonetheless, costs of other raw materials remained stable at low levels thanks to absences of pressures on global commodity prices.

Chart 2.19 Contribution to Producer Price Index



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand

Moreover, the government's plan to steadily increase LPG price to better reflect actual cost coupled with an upward adjustment in the FT following higher import costs of natural gas led to increases in domestic energy costs from 2014 Q4.

Inflationary pressure in the periods ahead

Overall inflationary pressure was not expected to raise concern. Despite some increase in cost pressure in line with the upward trend in domestic energy prices, domestic demand which continued to be weak along with declining inflation expectation of business sector would only result in minimal pass-through to prices.

(1) Cost pressure would likely increase, going forward, in line with continuous increase in LPG price between now and September 2014 together with increase in the FT for the May–

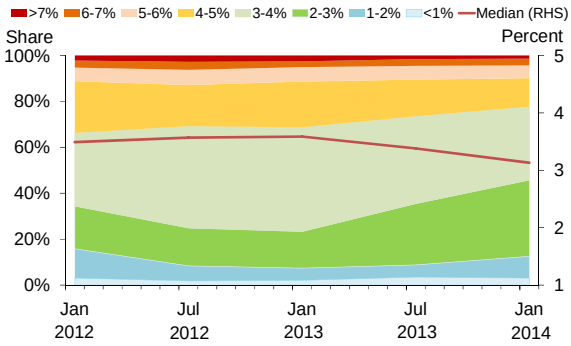
August 2014 round in tandem with higher energy costs. Nonetheless, the government's decision to maintain domestic diesel price and postpone LPG price hike for the transportation sector for the time being, would help relieve some pressure in the short-term. However, if price maintenance was lifted, it would pose an important upside risk to cost pressure.

(2) Demand pressure would continue to be low in line with the slowdown in economic activities, particularly private consumption and investment as businesses and households remained concerned about the prospect of domestic economic recovery along with the persistent political situation.

(3) Inflation expectations were expected to stabilize close to recent levels where businesses' 12-month-ahead inflation expectation in January 2014 stood at 3.13 percent. **(Chart 2.20)**

Concerning the short-term outlook of inflation, it was expected that both headline inflation and core inflation would continue to rise from current levels in line with gradually rising cost pressure. Nonetheless, pass-through to prices would be limited only to prepared food, judging from its higher-than-expected trend during the first two months of the year. Moreover, increases in prices of other goods and services in the Core Consumer Price Index basket were expected to be minimal given that overall cost of production still remained low. The low cost pressure coupled with short-term demand pressure which was expected to remain weak would make it difficult for businesses to raise prices of goods and services in periods ahead.

Chart 2.20 12-month ahead inflation expectation (January 2014)



Source: Bank of Thailand's Business Sentiment Survey

Dissolution of parliament: the impact on budget disbursement

The dissolution of parliament on December 9, 2013, followed by an unsuccessful attempt at forming a new government raised concerns among many about the impact of this political deadlock on future budget disbursement. This article was aimed at providing clarity on such issue as well as analysis of the extent of potential impact as follows.

(1) Fiscal year 2014 budget: Most items would be disbursed as usual but the central government budget's emergency provision along with some parts of capital expenditure might be affected.

Since the Annual Expenditure Budget Act B.E. 2557 (2014) was already enacted on October 11, 2013, disbursement according to the Act could therefore, be done as usual. However, because the Constitution of the Kingdom of Thailand B.E. 2550 (2007) granted the Acting Council of Ministers with only limited power[1/], some parts of budget disbursement might be affected, including:

1. **The central budget's emergency provision**. In normal circumstances, the Annual Expenditure Budget Act stipulates an overall amount of budget provision without details of individual items whereby any disbursement from this provision must be approved by the Council of Ministers as necessary on a case by case basis. In this case, items that were not approved by the Council of Ministers before the dissolution of parliament might not be available for disbursement given the Acting Council of Ministers' absence of power to approve new budget. An exception, however, could be granted with authorization from the Election Commission of Thailand according to Article 181(2) of the Constitution.

2. **Capital expenditure on new investment projects**, including additional borrowing limits and changes to expenditures formerly approved. These items must be approved by the

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[1/] Article 181 of the Constitution of Kingdom of Thailand B.E. 2550 (2007) stipulates that the outgoing Council of Ministers shall remain in office for performing duties until the newly appointed Council of Ministers takes office; but, in the case of the vacation of office under section 180 (2), the Council of Ministers and Ministers may perform duties only to the extent of necessity, subject to the following conditions:
(1) there shall not be taken any act which is, in nature, the exercise of the power to appoint or transfer Government officials holding permanent positions or receiving permanent salaries or officials of State agencies, State enterprises or undertakings of which the majority of shares are held by the State or dismiss such persons from the performance of duties or from office or instruct others to perform duties in replacement, unless upon prior approval by the Election Commission;
(2) there shall not be taken any act which has the effect of approving any disbursement of reserved funds available for cases of emergency or necessity, unless upon prior approval by the Election Commission;
(3) there shall not be taken any act which has the effect of approving any work or project or of creating binding commitment upon the subsequent Council of Ministers;
(4) there shall not be any exploitation of State resources or State personnel for any act having impacts on an election and there shall not be committed any act which is, in nature, a violation of prohibitions in accordance with the rules prescribed by the Election Commission.

Council of Ministers prior to disbursement[2/]. Thus, this part of capital expenditure also might not be available for disbursement given the Acting Council of Ministers' absence of power to bind new governments to its commitments according to Article 181(3) of the Constitution.

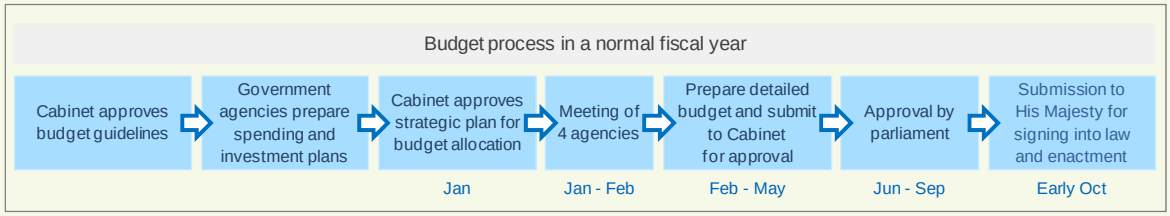
In this regard, during the first four months of fiscal year 2014, the government already disbursed a total of 1 trillion baht of budget, equivalent to 37.5 percent of the total budget, representing a 5.3 percent growth from to the same period last year. The BOT, therefore, assessed that the government would be able to disburse approximately 90.5 percent of the annual budget this fiscal year, overall.

Annual expenditure FY 2014 (billion baht)	Amount	Accumulated disbursement (Oct 2013–Jan 2014)	BOT's projections
Total expenditure	2,525	947	2,283
Rate (percent)		37.5	90.5
Current expenditure	2,097	822	1,993
Rate (percent)		40.3	95.0
Capital expenditure	428	125	286
Rate (percent)		25.8	67.0

Source: Bank of Thailand

(2) Fiscal year 2015 budget: issuance of the Annual Expenditure Budget Act B.E. 2558 (2015) would likely be delayed though the budget appropriation framework of the previous year could be used in the meantime.

In normal budget process, government offices, state enterprises and other government agencies prepare spending and investment plans according to pre-approved guidelines from the Council of Ministers. The process usually begins one month prior to submission of the plans for review. The budget review process is designed to provide a thorough, step-by-step examination to ensure that all expenditures are beneficial for the country and includes a review of the strategic plan on budget allocation prepared by the Bureau of the Budget and the National Economic and Social Development Board (NESDB), a joint review by four major economic institutions namely the Ministry of Finance, the Bureau of the Budget, the NESDB and the BOT, and submission to Parliament for approval. Once the entire process is completed, the budget can then be submitted to His Majesty King Bhumibol Adulyadej for signing into law before a new budget year can start in October.



Source: Annual budget calendar, Bureau of the Budget

[2/] According to Article 7 of the Rules on Debt Accumulation with Binding Effect over One Fiscal Year B.E. 2534 (1991) as amended in B.E. 2549 (2006)

Given the current circumstance in which a new government could not be formed successfully and the budget process could not officially begin, it would be possible that enactment of the Annual Expenditure Budget Act B.E. 2558 (2015) could be delayed or unable to meet the budget deadline at the beginning of October.

Notwithstanding the potential delay of the new budget, provisions under the Constitution[3/] and the Budget Procedures Act B.E. 2502 (1959) [4/] would make it possible to use annual appropriations approved in the previous year's budget in the meantime, as follows:

1. Current expenditure–according to the appropriations approved in the previous year's budget for salary payments
2. Capital expenditure–only items in connection to the previous year's budget
3. Subsidies–only items that were always subsidized in the past
4. Transfers to local governments–according to the same rules as 1–3

Historical records revealed that whenever there was a delay in the budget, disbursement would be lower than target. For example, in fiscal years 2007 and 2012, announcements of the Annual Expenditure Budget Act were delayed by 3 and 4 months, respectively. Disbursement rates, thus, stood at 93.6 and 90.3 percent, respectively.

Dissolution of parliament	Issuance of next fiscal year's budget	Disbursement rate (%)
February 2006	Jan 2007 (3-month delay)	93.6
May 2011	Feb 2012 (4-month delay)	90.3

In summary, the impact of the dissolution of parliament on budget disbursement for fiscal year 2014 was judged to be minimal though some parts of the central budget and capital expenditure might be affected. However, it was expected that issuance of next fiscal year's budget would be delayed. Nonetheless the annual appropriation framework approved in the previous year's budget could still be used in the meantime.

[3/] Article 166 of the Constitution of Kingdom of Thailand B.E. 2550 (2007) stipulates that the expenditure estimates of the State shall be made in the form of an Act. If the Annual Appropriations Act for the following fiscal year is not enacted in time, the law on annual appropriations for the preceding fiscal year shall apply for the time being.

[4/] Article 16 of the Budget Procedures Act B.E. 2502 (1959) states that in case Annual Budget Expenditure Act could not be issued on time for the new fiscal year, the budget that has duly passed shall be used in continuation for the time being as per principle and standard and conditions prescribed by the Director by consent of the Prime Minister.

Monetary and Financial Stability

3. Monetary and Financial Stability

Thailand's overall economic and financial stability continued to be well maintained. Money market interest rates moved in synchronization with the policy rate. Financial strength of the financial institutions system remained sound. Meanwhile, non-financial institution sectors began to be affected by the economic slowdown such that households' debt service burden increased and corporate profitability edged down slightly from the first half of the year. Moreover, concerns over the Fed's tapering of Quantitative Easing (QE) measures caused financial markets to become highly volatile in 2013 Q4, particularly the equity market.

Important risks that needed to be monitored in the periods ahead included 1) uncertainty in the global economy that could lead to sudden capital flight and financial market volatility; 2) slowdown in the domestic economy that could affect household income and corporate profitability, thereby leading to deteriorating debt servicing ability and financial institutions' loan quality; and 3) prolonged political situation that could put a dent in private sentiment and households' confidence.

3.1 Financial markets

Money market interest rates adjusted downwards in tandem with the policy rate while movement of government bond yields was volatile due to demand for bond investment from both domestic and foreign investors that resulted from the domestic political situation and expectation of QE tapering.

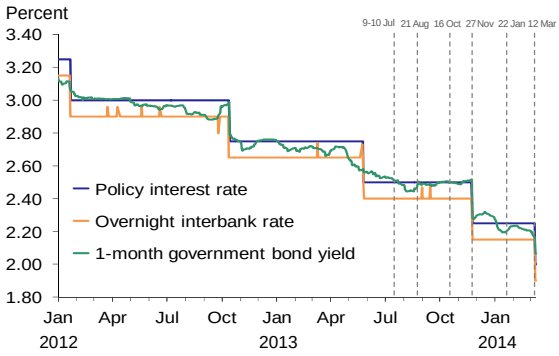
Money and bond markets

During the second half of 2013, short-term money market rates continued to stabilize close to the policy rate which the MPC decided to maintain at 2.50 percent per annum at the July 10, August 21 and October 16, 2013 meetings. Then, short-term money market rates adjusted downwards

following the MPC's decision to decrease the policy rate from 2.50 to 2.25 percent per annum at the November 27, 2013 meeting. At the end of 2013, the overnight interbank lending rate and the one-month government bond yield stood at 2.15 and 2.30 percent per annum, respectively.

During January–March, 2014, short-term money market rates remained close to the policy rate prior to a downward adjustment in accordance with the MPC's decision to decrease the policy rate again, from 2.25 to 2.00 percent per annum at the March 12, 2014 meeting. In this regard, the overnight interbank lending rate edged down to 1.90 percent per annum on March 13, 2014. The one-month government bond yield also declined owing to (1) expectation of policy rate decreases at the January and March, 2014 MPC meetings; (2) higher demand for bond investment from domestic and foreign investors; and (3) a policy rate decrease in March. As such, the one-month government bond yield, thus, decreased to 2.06 percent per annum in March 13, 2014. **(Chart 3.1)**

Chart 3.1 Money market short-term interest rates



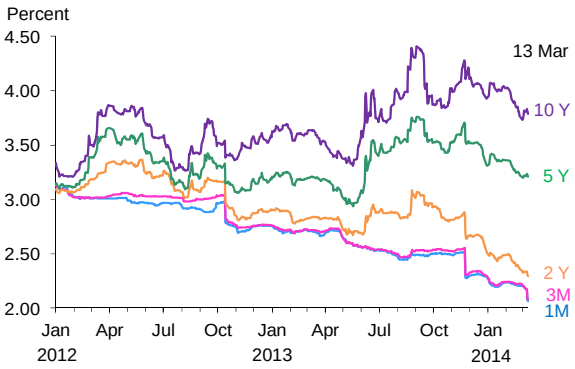
Sources: Bank of Thailand and the Thai Bond Market Association (ThaiBMA)

During the second half of 2013, short-term government bond yields declined from the first half of the year in line with direction of the policy interest rate and higher demand for bond investment from both domestic and foreign investors. Meanwhile, medium- to long-term government bond yields were slightly volatile and edged up in tandem with the U.S. treasury yields following financial markets' anticipation of a sooner-than-expected QE tapering by the Fed which resulted in asset sell-offs by foreign investors. However, since November, medium- to long-term government bond yields somewhat trended downwards following the MPC's decision to decrease the policy rate on November 27, 2013, along with higher demand for bond investment by

domestic and foreign investors that was in part driven by concerns over potential reduction in new supply of government bonds following the dissolution of parliament coupled with possible delays in investment in large infrastructure development projects.

From January to March 2014, government bond yields became volatile and adjusted downwards overall in tandem with financial markets' expectation of a policy rate decrease by the MPC together with higher demand for bond investment from domestic investors given their preference for safer assets due to concerns over the ongoing economic and political uncertainties. Asset sell-offs by foreign investors slowed down in line with greater stability in the global financial markets after a period of concerns over fragilities in the fundamentals of emerging market economies subsided. In this regard, the MPC's decision to decrease the policy rate on March 12, 2014 only caused government bond yields to decline marginally since the decision was already priced in by most investors. **(Chart 3.2)**

Chart 3.2 Government bond yields

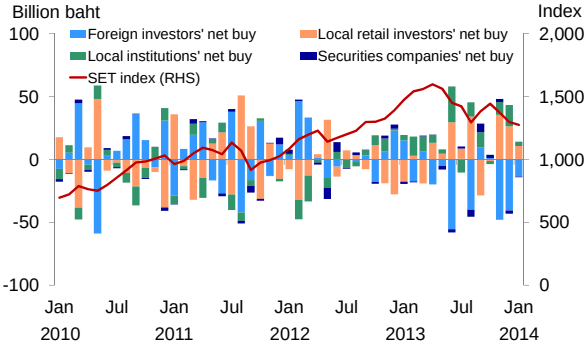


Sources: The Thai Bond Market Association (ThaiBMA)

Equity market

From 2013 Q4 to January 2014, movement in the Stock Exchange of Thailand (SET) index was rather volatile due to both domestic and external factors. On the domestic front, there was political uncertainty while on the external front, there was the Fed's QE tapering which affected investors' confidence, leading to continuous asset sell-offs. As a result, volatility in the SET index during 2013 Q4 remained high at 19.8 percent per year. Meanwhile, the SET index also trended downwards continuously to reach a new low record of 1,225 points on January 3, 2014 while foreign investors' net sell during October

Chart 3.3 The Stock Exchange of Thailand index
(As of January, 2014)



Source: The Stock Exchange of Thailand

2013–January 2014 registered as high as 101 billion baht. **(Chart 3.3)**

Important risks to be monitored in the periods ahead included (1) ongoing risk of capital flow volatility which could affect Thailand's equity market; and (2) domestic political uncertainty which could put a dent in investors' confidence.

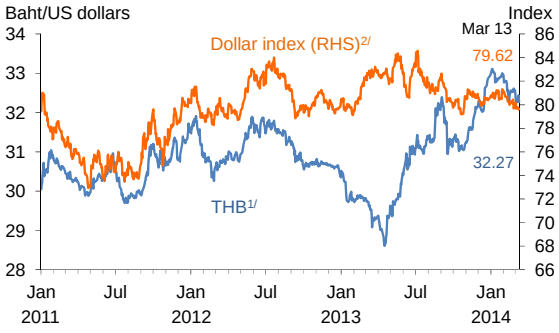
Foreign exchange market

Since February 2014, the Thai baht resumed stability against the U.S. dollar after a year of volatility due to global financial conditions. On March 13, 2014, the Thai baht stood at 32.27 baht per U.S. dollar, appreciating from the end of 2013 by 1.8 percent **(Chart 3.4)** owing to (1) relatively well contained political situation despite prolonged conflicts; (2) depreciation in the U.S. dollar following weak outturn in economic data; and (3) improvement in Asian economies, which led investors to regain confidence and alleviate concerns about potential emerging market crisis contagion.

In this regard, factors that caused the Thai baht to depreciate continuously during the latter part of 2013 included speculation on the Fed's QE tapering following announcements of more promising economic data, and emerging fragilities in emerging market economies. These two factors together led to capital outflows from regional economies including Thailand. Another notable factor underlying weakness in the Thai baht was the domestic political situation.

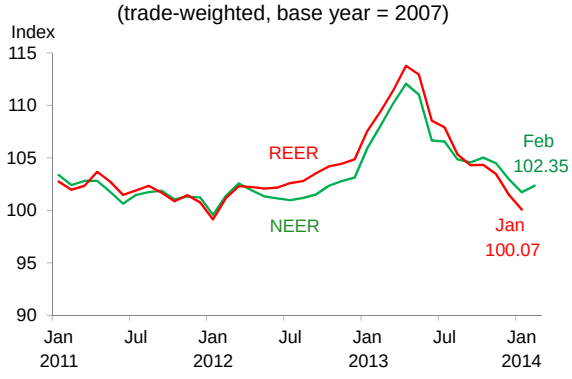
In February, the Nominal Effective Exchange Rate (NEER) registered at 102.35, appreciating from last month by 0.6 percent **(Chart 3.5)** mainly from the strengthening of the Thai baht against the Chinese yuan, the U.S. dollar, and the euro. Such

Chart 3.4 Exchange rate and trade-weighted dollar index



Sources: 1/ Bank of Thailand
2/ Bloomberg

Chart 3.5 NEER and REER (trade-weighted, base year = 2007)



Source: Bank of Thailand

development was due to the slowdown in China's manufacturing sector coupled with the euro area's weakening economic outlook following the decline in inflationary pressure. Subsequently, financial markets began to expect further monetary easing by the European Central Bank (ECB). Meanwhile, the Real Effective Exchange Rate (REER) stood at 100.07 in January, depreciating by 1.4 percent from the previous month owing to the depreciation in the NEER. For 2013 overall, it was found that the NEER and REER both assumed weakening trends towards the second half of the year, reflecting improvement in Thailand's price competitiveness compared to trading partners.

3.2 Financial institutions

During the second half of 2013, deposit rates on both normal time deposits and special deposit products as well as loan rates of commercial banks adjusted downwards in line with the trend in the policy rate along with waning competition for deposits which was consistent with financial institutions' plans to reduce lending during the economic slowdown.

Interest rates, deposits and credits

Commercial banks' interest rates on both deposits and loans adjusted downwards during the second half of 2013 following the MPC's decision to decrease the policy rate in November 2013. At the end of 2013, interest rates on the 12-month deposits of the four largest commercial banks[1/] averaged at 2.23 percent per annum, declinding from 2.40 percent per annum recorded

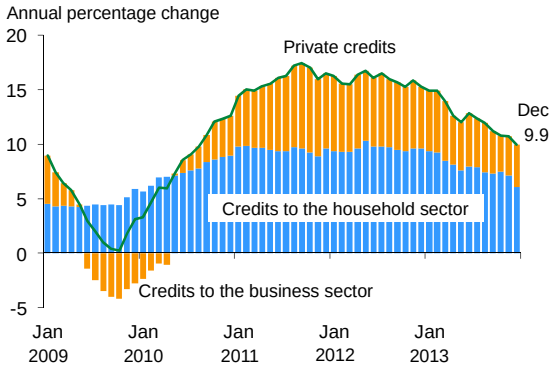
[1/]BBL, TMB, KBank and SCB

at the end of the first half of the year. Towards the end of the year, many commercial banks and Specialized Financial Institutions (SFIs) also reduced interest rates on existing special deposit products as a result of waning competition for deposits, which was consistent with financial institutions' plans to reduce lending gradually during the economic slowdown.

Meanwhile, the average MLR of the four largest commercial banks at the end of 2013 stood at 6.84 percent per annum, declining from 7.00 percent per annum recorded at the end of the first half of the year.

Between January and February 2014, many commercial banks continued to reduce interest rates on normal time deposits and existing special deposit products and refused roll-overs of products falling due, in line with waning competition for deposits. At the end of February 2014, the interest rates on the 12-month deposits of the four largest commercial banks averaged at 2.10 percent per annum while the average MLR edged up slightly from the end of 2013 to 6.88 percent per annum owing to the increase in MLR by one commercial bank to match the levels of the other three, after it had previously decreased its MLR by more than the others.

Chart 3.6 Other depository corporations' private credits



Source: Bank of Thailand

Private credits slowed down continuously in the second half of the year. At end-December 2013, private credits expanded by 9.9 percent from last year, moderating from 12.9 percent at the end of the first half of the year **(Chart 3.6)** owing to the slowdown in the economy together with financial institutions' more stringent lending practice. Consumer credits trended downwards partly due to the high level household debt. In particular, auto leasing slowed down substantially

after deliveries under the first car tax rebate scheme was completed. At the same time, corporate credits moderated in line with private investment. Moreover, the BOT's Senior Loan Officer Survey at end-December 2013 revealed that commercial banks would raise their lending standards further, especially for SMEs, auto leasing and other personal loans.

Deposits and bills of exchange (B/E) moderated in the second half of the year after high acceleration in the first half. At the end of 2013, deposits and B/E grew by 7.5 percent from last year, slowing down from 10.4 percent registered at the end of the first half. The slowdown was observed in deposits of both corporate and households in line with financial institutions' plans to reduce lending during the economic slowdown, which resulted in waning competition for deposits. In addition, given greater interest in savings alternatives towards the end of the year, for example, mutual funds and bonds, the Loan to Deposit and B/E ratio continued to increase from 95.3 at the end of 2013 Q2 to 96.6 at year-end. Looking ahead, it was expected that competition for deposits would continue to lessen owing to various factors, namely financial market conditions, corporate bond issuance by the private sector and demand for credits, albeit subject to the economic conditions.

Stability of financial institutions

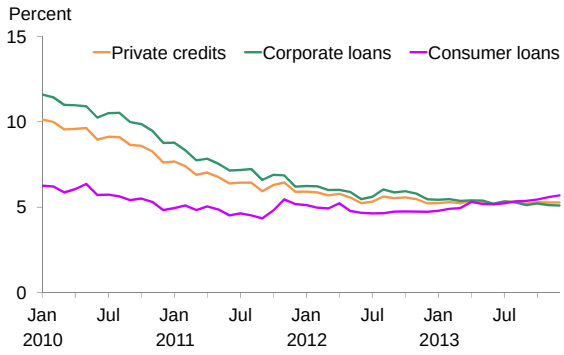
Stability of financial institutions in the second half of 2013 remained sound. Commercial banks' profitability declined slightly from the second quarter owing to a special item arising from one commercial bank's profits from the sale of its subsidiary. The capital adequacy ratio

remained high. Meanwhile, overall loan quality slightly deteriorated due to auto leasing.

Commercial banks continued to pose strong profitability in the second half of 2013 despite a continuous decrease in net profit from 58.1 billion baht at the end of 2013 Q2 to 54.4 and 50.2 billion baht at the end of 2013 Q3 and Q4, respectively. **(Table 3.1)** Excluding the special item on the sale of a commercial bank's subsidiary, net profit of commercial banks in 2013 Q3 would increase by 6.1 billion baht from 2013 Q2 before decreasing by 4.16 billion baht in 2013 Q4 owing to rising operating cost along with higher provisioning for loss and doubtful loans. Meanwhile, commercial banks continued to be well capitalized in spite of a small decrease in the capital adequacy ratio from 16.5 percent in 2013 Q3 to 15.7 percent in 2013 Q4. Moreover, overall loan quality remained sound though some deterioration in auto leasing was observed as reflected by an increase in delinquency and NPL ratio from 5.2 percent in 2013 Q2 to 5.4 and 5.7 percent in 2013 Q3 and Q4, respectively. In contrast, delinquency and NPL ratio of overall private credits remained stable from the first half of the year at 5.3 percent. **(Chart 3.7)** Nonetheless, liquidity tightened slightly in the second half of the year as reflected by the ratio of Loan to Deposits and B/E that increased continuously from 2013 Q2, owing to the fact that deposits expanded less than credits.

Looking ahead, factors to keep abreast included the impact from the slowdown in the economy along with the prolonged political situation, which could affect liquidity of the private sector and subsequently worsen debt servicing ability and loan quality.

Chart 3.7 Delinquency and NPL ratio



Source: Bank of Thailand

3.3 Non-financial sectors

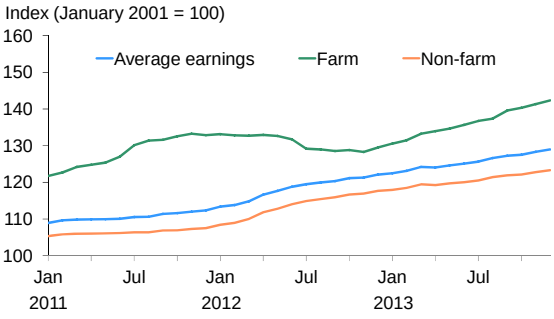
Overall stability in non-financial sectors was well maintained though risks increased in some sectors, namely household, corporate and real estate sectors. Development in these sectors would thus, require close monitoring in the periods ahead, especially from the economic slowdown which could affect household income and corporate profitabiity and subsequently debt servicing ability. Moreover, domestic political uncertainty could also undermine private sentiment, if problems persisted.

Household sector

Overall stability of the household sector remained intact thanks to the slowdown in household debt. However, the slowing economy could potentially affect income and debt servicing ability of households, going forward.

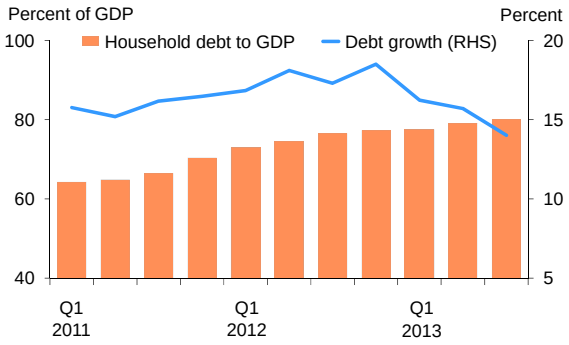
Household income improved as reflected by average earnings for both farm and non-farm sectors that trended upwards throughout 2013. **(Chart 3.8)** Employment conditions remained favorable such that the average unemployment rate stood at a low level of 0.7 percent in 2013. Furthermore, after strong acceleration in prior periods, household debt declined continuously as the first car tax rebate scheme gradually expired and households became more cautious in their spending and debt accumulation. Financial institutions also adopted more stringent practice in lending. In this regard, household debt (financial institutions' loans to households) at the end of 2013 Q3 expanded by 14.0 percent, slowing down from 15.7 percent in the previous quarter. However, growth in household debt still remained

Chart 3.8 Household income*



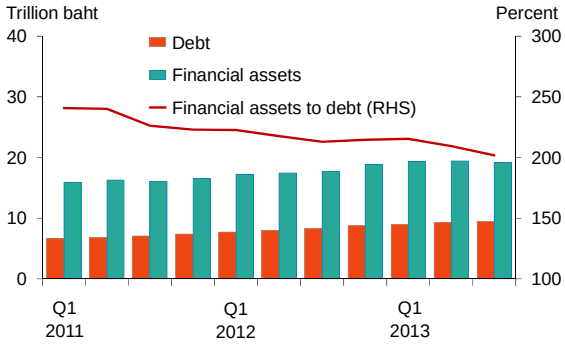
Note: *seasonally adjusted, real income (12-month moving average)
Sources: Office of Agricultural Economics, National Statistics Office, calculations by Bank of Thailand

Chart 3.9 Ratio of household debt* to GDP



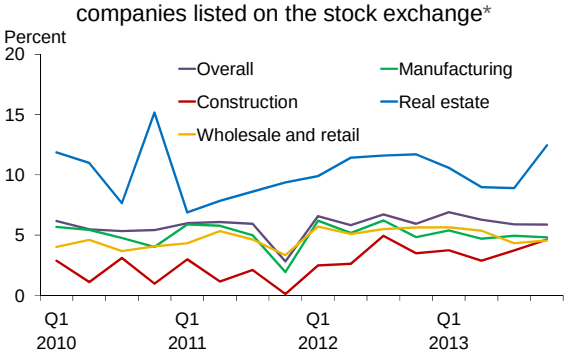
Note: *Loans to households from financial institutions
Source: Bank of Thailand

Chart 3.10 Ratio of financial assets to debt*



Note: *Loans to households from financial institutions
Source: Bank of Thailand

Chart 3.11 Net Profit Margin (NPM) of non-financial companies listed on the stock exchange*



Note: *Median
Sources: Stock Exchange of Thailand and calculations by Bank of Thailand

higher than that in GDP, in part as a result of high competition in the market for consumer loans. Given the more-than-expected slowdown in the economy, the ratio of household debt to GDP thus increased slightly, from 79.2 at the end of 2013 Q2 to 80.1 at the end of 2013 Q3. **(Chart 3.9)** In addition, household liquidity also tightened somewhat as reflected by the ratio of financial assets to debt of households which continued to edge down from the previous quarter to stand at 201.7 at the end of 2013 Q3. **(Chart 3.10)**

Debt servicing ability of households worsened marginally as reflected by an increase in delinquency ratio of consumer loans from 5.4 percent at the end of 2013 Q3 to 5.7 percent at the end of 2013 Q4, owing chiefly to auto leasing. As a result, household income and debt servicing ability would warrant close monitoring in the periods ahead, especially given households' already high indebtedness that was due to accelerated borrowings in previous periods. In this connection, any potential negative development could therefore lead to instability in the household sector, going forward.

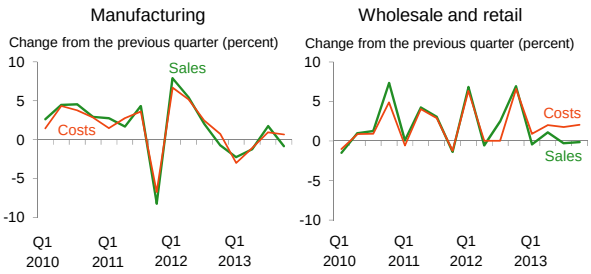
Corporate sector

The global and domestic economic slowdown started to affect corporate profitability slightly, in particular, the manufacturing, wholesale and retail sectors, causing some deterioration in the overall debt servicing ability. Nonetheless, retained earnings, to certain extent, remained a cushion for negative impact on the corporate sector.

During the second half of 2013, Net Profit Margin (NPM) of the corporate sector registered at 5.9 percent, declining slightly from 6.3 percent in 2013 Q2. **(Chart 3.11)** Although the domestic

economic slowdown began to have some impact on the corporate sector, overall stabiity continued to be intact with retained earnings, debt servicing ability and liquidity remaining sound. The interest coverage ratio increased from 3.1 in 2013 Q2 **(Chart 3.12)** to 4.0 and 4.2 in 2013 Q3 and Q4, respectively, partly as a result of the slowdown in debt accumulation. Meanwhile, the Current Ratio (CR) edged up to 1.6 in 2013 Q4. However, closer examination revealed that fragilities built up in some sectors in 2013 H2 as a result of the economic slowdown, for example, the manufacturing and the wholesale and retail sectors, with the latter also being significantly affected by higher production costs. In this regard, the Debt to Equity (D/E) ratio of wholesale and retail sector started to climb from 0.8 in 2013 Q2 to 0.9 and 1.0 in 2013 Q3 and Q4, respectively while the Current Ratio deteriorated from 1.5 in 2013 Q3 to 1.3 in 2013 Q4.

Chart 3.12 Greater vulnerabilities in sales and costs*



Note: *Median

Sources: Stock Exchange of Thailand and calculations by Bank of Thailand

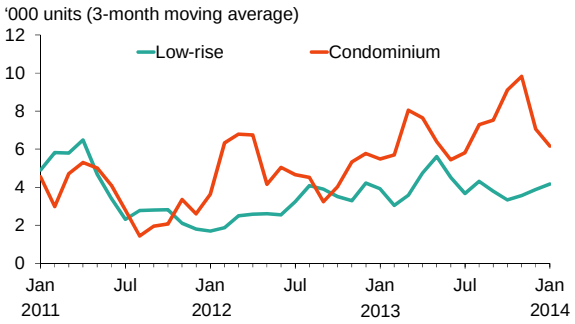
Looking ahead, factors that continued to warrant close monitoring included the impact of the economic slowdown which could impair corporate profitability, particularly in small and medium enterprises, thereby deteriorating debt servicing ability.

Real estate sector

Stability in the real estate sector remained sound overall where markets in Bangkok and vicinities began to cool down in tandem with economic development along with deteriorated consumers' and businesses' sentiments resulting from the short-term political situation.

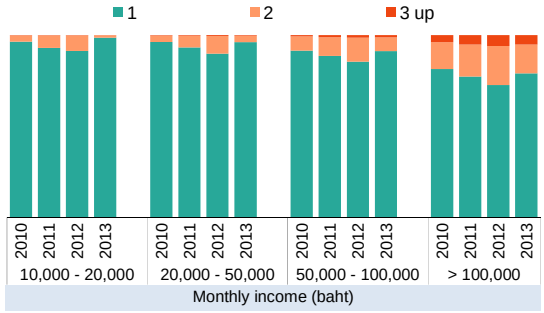
Overall, real estate market conditions in 2013 Q4 improved from the previous quarter, especially in condominiums thanks to new Expo to stimulate demand. However, from December onwards, the market started to slowdown both in

Chart 3.13 Number of new launches in Bangkok and its vicinities*



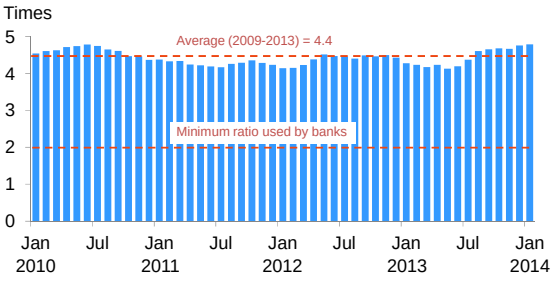
Note: *3-month moving average (seasonally adjusted data as of January 2014)
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Chart 3.14 Commercial banks' mortgages classified by number of properties held by borrowers*



Note: * Mortgages for new and pre-owned properties from Jan 2008–Dec 2013, calculated using a rolling window of 3 years and then comparing the number of borrowers in the third year of each window
Source: Bank of Thailand

Chart 3.15 Debt Service Coverage Ratio (DSCR)



Note:
(1) DSCR = monthly income/monthly debt payment (principal and interest)
(2) Calculated for new mortgages in Bangkok and its vicinity
(3) 50th percentile (3-month moving average)
(4) Includes only debt services for new mortgages by commercial banks
Source: Bank of Thailand

terms of demand and supply following strong rises in prior periods. Factors that contributed to such market cooldown were the economic slowdown and the domestic political situation which caused consumers to postpone purchasing decisions to future periods. Development in new supply was in line with demand, in particular, condominiums though supply of low-rise residential properties increased **(Chart 3.13)** as developers chose to diversify investment given that construction cost of low-rise residential properties could be managed more easily than condominiums.

Overall stability in the real estate sector was well maintained. The signal of market slowdown led to adjustments by all stakeholders to handle upcoming risks. Consumers were more cautious in their spending as reflected by reduction in the proportion of second home mortgages **(Chart 3.14).** New launches were postponed by developers, especially condominiums whereby the domestic political uncertainty was also responsible for part of this delay. In addition, there were signs of more stringent mortgage approvals as reflected by the Debt Service Coverage Ratio (DSCR) of 4.8 in January 2014 which was higher than the average of the last five years of 4.4. **(Chart 3.15)**

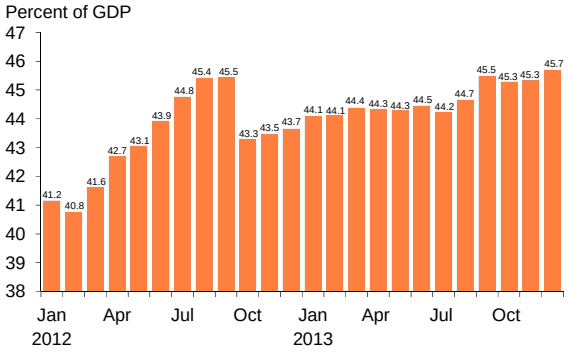
The real estate market slowdown led to some easing in price pressure as well as construction cost. In spite of well maintained stability overall, a factor that required close monitoring in the next periods included the economic slowdown which could negatively affect consumers' purchasing power and therefore, demand for real estate.

Fiscal sector

Fiscal sector stability continued to be sound though public debt to GDP edged up slightly in 2013 Q4. Looking ahead, key risks to keep abreast included implementation of the rice mortgage scheme and lower-than-targeted revenue collection in fiscal year 2014.

Fiscal sector stability continued to be sound despite the marginal increase in public debt to GDP in 2013 Q4. As of December 2013, public debt to GDP stood at 45.7 percent, rising from 45.5 percent at the end of the previous quarter. **(Chart 3.16)** Going forward, key risks to be mindful of included (1) the government's quasi-fiscal activities, particularly the rice mortgage scheme; (2) lower-than-targeted revenue collection owing the economic slowdown; and (3) the political situation that if persisted, could affect Thailand's credit rating, thus causing financing costs to hike up.

Chart 3.16 Public debt to GDP



Note: Fiscal year
Source: Public Debt Management Office

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2012	2013	2013						2014
			Q1	Q2	Q3	Q4	Nov	Dec	Jan
1. Financial markets sector									
Bond market									
Bond spread (10 years–2 years)	0.4	1.0	0.8	0.8	1.1	1.2	1.3	1.4	1.5
Equity market									
SET Index	1,391.9	1,298.7	1,561.1	1,451.9	1,383.2	1,298.7	1,371.1	1,298.7	1,274.3
Actual volatility (SET Index)[1/]	12.9	19.6	11.7	20.8	26.1	19.8	20.0	15.7	23.0
Price to earnings ratio (times)	14.1	16.7	17.6	17.5	15.5	15.7	15.7	15.1	14.2
FX market									
Actual volatility (baht) (%annualize)	4.6	6.2	4.31	7.35	7.82	4.4	3.5	4.5	3.4
Nominal effective exchange rate (NEER)	101.6	106.9	108.0	109.9	105.3	104.2	104.5	103.0	101.7
Real effective exchange rate (REER)	102.6	107.5	109.5	111.7	105.9	103.1	103.4	101.5	100.1
2. Financial institutions sector[2/]									
Minimum lending rate (MLR)[3/]	7.0	6.8	7.0	7.0	7.0	6.8	6.9	6.8	6.9
12-month fixed deposit rate[3/]	2.5	2.2	2.4	2.4	2.5	2.2	2.4	2.2	2.2
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.3	15.7	15.9	15.7	16.5	15.7	16.1	15.7	n.a.
Earnings and profitability									
Net profit (billion baht)	173.8	214.9	52.1	58.1	54.4	50.2			
Return on assets (ROA)	1.25	1.33	1.40	1.25	1.40	1.27			
Liquidity									
Loan to deposit and B/E	93.1	96.6	93.4	95.3	95.5	96.6	95.0	96.6	n.a.
3. Household sector									
Debt to financial assets	214.6	n.a.	215.3	209.4	201.7	n.a.			
NPL and delinquency ratio (%)									
Thai commercial banks :									
Consumer loans	4.7	5.7	4.9	5.2	5.4	5.7			
Mortgage loans	3.7	3.9	3.7	3.7	3.7	3.9			
Auto leasing	7.4	9.7	7.7	8.5	8.8	9.7			
Credit cards	3.7	4.7	4.4	4.6	4.9	4.7			
Other personal loans	3.5	4.6	3.9	3.9	4.2	4.6			
4. Non-financial corporate sector [4/]									
Operating profit margin (%)	6.3	6.2	6.9	6.3	5.9	5.9			
Debt to equity ratio (times)	0.8	0.7	0.8	0.8	0.8	0.7			
Income coverage ratio (times)	3.5	3.7	3.6	3.1	4.0	4.2			
Current ratio (times)	1.5	1.6	1.5	1.5	1.5	1.6			

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability (cont.)

Indicators	2012	2013	2013						2014
			Q1	Q2	Q3	Q4	Nov	Dec	Jan
5. Real estate sector									
3-month moving average of the number of approved mortgages from banks (Bangkok and its vicinity)	76,052	73,532	19,777	17,274	15,999	20,509	6,320	7,006	5,001
Single-detached and semi-detached houses	22,914	19,363	5,535	4,571	3,775	5,482	1,721	1,871	1,205
Townhouses and commercial buildings	26,296	26,151	7,724	6,041	5,265	7,121	2,167	2,589	1,940
Condominiums	26,842	28,018	6,518	6,635	6,959	7,906	2,432	2,546	1,856
3-month moving average of the number of new openings (Bangkok and its vicinity)	96,275	133,293	29,776	34,343	32,359	36,778	13,388	10,946	10,327
Single-detached and semi-detached houses	14,449	16,904	4,651	4,286	4,057	3,909	1,223	1,385	1,450
Townhouses and commercial buildings	21,584	31,052	5,895	10,580	7,714	6,863	2,339	2,491	2,715
Condominiums	60,242	85,337	19,230	19,477	20,624	26,006	9,826	7,070	6,162
Housing price index[5/]									
Single-detached houses (including land)	106.2	112.2	107.8	109.4	113.7	112.2	113.1	112.2	112.4
Townhouses (including land)	113.2	124.2	118.3	118.6	123.7	124.2	124.8	124.2	124.3
Condominiums	114.8	119.7	118.7	117.6	121.7	119.7	120.9	119.7	120.0
Land	117.2	132.2	123.2	120.6	124.2	132.2	129.7	132.2	133.1
6. Fiscal sector									
Public debt to GDP (%)	43.7	45.8	44.4	44.5	45.5	45.8	45.3	45.7	45.8

[1/] Daily volatility (using exponentially weighted average method)

[2/] Based on data of all commercial banks

[3/] Average value of 4 largest Thai commercial banks

[4/] Only listed companies on SET (median)

[5/] Based on data of new approvals by commercial banks using hedonic regression method (January 2010 = 100)
(Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change)

Developing the new Nominal Effective Exchange Rate (NEER)

The Nominal Effective Exchange Rate (NEER) and the Real Effective Exchange Rate (REER) are two indicators of movement of the baht compared to trading partners' currencies, which are used to assess the country's price competitiveness along with its impact on the overall economy.

Construction of the NEER[1/] to represent movement of the baht compared to other currencies according to their relative importance in terms of trade requires selection of countries and assignment of weights to the chosen currencies that reflect the structure of trade between Thailand and each chosen country. Additionally, the weights must be revised consistently in order to ensure that the structure of trade can be most accurately proxied in each time period.

The BOT, thus, proceeded to revise the weights used in the construction of the NEER and REER in order to capture the true structure of trade as follows.

(1) Used trade data in 2012 instead of 2007 as weights for the calculations of the NEER and REER from 2012 onwards. Given that Thailand's and the global economies became quite stable after the global economic crisis in 2008, it was assessed that trade data in 2012 would be appropriate representations of today's structure of trade.

(2) Used 2012 as base year instead of 2007 to ensure consistency with year of trade data used.

(3) Improved data filtering method in the calculation of trade value

(3.1) Included export and import value of services in the calculation of trade weights for Thailand, given that exports of services, particularly tourism, became increasingly more important to the Thai economy compared to previous periods.[2/]

(3.2) Excluded export and import value of gold from Thailand's trade data since gold were more frequently regarded as investment goods rather than traded goods, therefore, it did not reflect the country's competitiveness. As a result, significant increases in export and import volume of gold in recent periods may, in fact, distort actual structure of trade.

(3.3) Excluded data on re-exports and re-imports[3/] to avoid double counting. For example, in case of re-exports, if a country was in the trade chain like Hong Kong and Singapore, the country at the beginning of the chain would be regarded as the true exporter of such re-export goods.

[1/] For further information, please refer to Focused and Quick (FAQ) No. 86 "Getting to know the NEER" (http://www.bot.or.th/Thai/EconomicConditions/Publication/FAQ_documents/FAQ_86.pdf)

[2/] Nonetheless, data limitation on the services sector in other countries made it impossible to fully include such data into the calculation of trade weights.

[3/] Reasons for re-import include but would not be limited to return of faulty goods, recall of cancelled orders, payment default or trade barrier in importing countries.

(4) For competition in a third market between Thailand and an oil-exporting country, namely Saudi Arabia and United Arab Emirates, a "zero" weight was assigned since Thailand did not compete directly with these countries in that third market.

The revisions outlined above resulted in adjustments in the number of countries and trade weights of Thailand's important trading partners, as follows.

(1) The number of trading partners' currencies in the calculation of the NEER increased from 23 to 25 (Table 1) from the addition of Myanmar, Laos and Cambodia, owing to their rising importance in trade which was expected to continue in the future. Moreover, Canada was removed from the calculation since the proportion of its trade and extent of competition with Thailand declined significantly. In addition, given Estonia's adoption of the euro (making it the 17th country to join the single currency), the total number of countries that were included in the calculations of trade weights increased from 38 to 41 countries.

Comparison of the number of countries
and currencies included in the calculation of the NEER

Items	NEER-23	NEER-25
Year of trade weights calculation	2007	2012
Base year	2007	2012
Number of currencies	23	25
Number of countries	38 (16 euro adopters)	41 (17 euro adopters)
Percent of total trade value	84.8	90.7

(2) The weight of each currency was changed. In particular, while China received a larger weight, Japan and the euro area were assigned smaller weights due to their declining roles as Thailand's trading partners. This observation was consistent with recent dynamics in international trade.

The BOT would begin using the new NEER and REER in March 2013 while data would be revised backwards to January 2012. The base year would also be changed to 2012, which was also the year of data used in the calculations of trade weights. Moreover, the BOT would revise the weights along with calculation methods every three years in order to ensure that the indicators could capture the true structure of trade in line with changing international trade dynamics.